

Senior High School



# Lesson Exemplar in General Mathematics

Quarter 3

Unit

8

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Lesson Exemplar for General Mathematics  
Quarter 3: Unit 8

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Republic of the Philippines  
**Department of Education**  
BUREAU OF LEARNING DELIVERY

| LESSON EXEMPLAR |                            |                       |                        |
|-----------------|----------------------------|-----------------------|------------------------|
| Learning Area:  | <b>GENERAL MATHEMATICS</b> |                       | Grade Level: <b>11</b> |
| Semester:       | <b>SECOND</b>              | Quarter: <b>THIRD</b> | Unit: <b>8</b>         |

| I. LEARNING GOALS                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>Content Standard</b>             | <p>The learners demonstrate knowledge and understanding of practical applications of measurements.</p> <p><i>Contextualized:</i><br/><b>The learners demonstrate knowledge and understanding of how measurement concepts are applied in real-life situations such as estimating and measuring lengths, areas, volumes, and capacities in activities at home, school, and the community.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Performance Standard</b>         | <p>By the end of the quarter, the learners are able to apply trigonometry and geometric concepts to estimate measurements and construct three-dimensional objects, understanding their shapes, and volumes for practical uses like product packaging, furniture and construction.</p> <p><i>Contextualized:</i><br/><b>By the end of the quarter, the learners are able to apply trigonometric and geometric concepts to solve real-life measurement problems such as estimating dimensions, designing, and constructing three-dimensional objects used in daily life (e.g., creating product packaging, designing furniture, planning simple structures, or measuring spaces for construction purposes) demonstrating accuracy, creativity, and practical understanding of shapes and volumes.</b></p> |
| <b>Learning Competencies</b>        | <p>The learners...</p> <ol style="list-style-type: none"><li>1. estimate and measure lengths and distances (e.g. floor dimensions measuring fabric for clothing).</li><li>2. calculate the area and perimeter of residential spaces (e.g., room renovations, garden layouts).</li><li>3. calculate the volume and capacity of three-dimensional objects (e.g., water tanks, rice storage containers, balik-bayan box).</li><li>4. apply volume and capacity in cost estimation (e.g., amount of flour needed for bread production, paint needed for walls).</li></ol>                                                                                                                                                                                                                                   |
| <b>II. REFERENCES and MATERIALS</b> | <p><b>References</b></p> <p>Jaen, C et al (2011), Trademarks of StudyPad, Inc. <i>Measurement – Definition, Examples, Practice Problems, FAQs.</i><br/><a href="https://www.splashlearn.com/math-vocabulary/measurements/measurement">https://www.splashlearn.com/math-vocabulary/measurements/measurement</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

Siyavula (2025). *Estimating and measuring length and distance*.  
<https://www.siyavula.com/read/za/mathematical-literacy/grade-10/measuring-length-weight-volume-and-temperature/05-measuring-length-weight-volume-and-temperature-02>

Resource Center for CareerTech Advancement (2019). *Math Series Measurement: Measuring Lengths and Distances*  
<https://oklahoma.gov/content/dam/ok/en/careertech/educators/resource-center/tools-for-teachers-and-trainers/measuring-lengths-and-distances.pdf>

Oxford University Press (2025). *Units of measurement and applications*.  
[https://www.oup.com.au/\\_\\_data/assets/pdf\\_file/0029/58295/Oxford-Insight-Maths-General-Preliminary-ch2-Units-of-measurement-and-applications.pdf](https://www.oup.com.au/__data/assets/pdf_file/0029/58295/Oxford-Insight-Maths-General-Preliminary-ch2-Units-of-measurement-and-applications.pdf)

Mary Palmer (in *Timeless Landscape Design*) cited via Dargan Garden Design. "The Perimeter: Extending Your Living Spaces to the Outdoors." Dargan Garden Design, April 7, 2014. Retrieved from <https://dargan.com/2014/04/07/the-perimeter-extending-your-living-spaces-to-the-outdoors/>

Hake, S. (2007). *Saxon math course 1*. Houghton Mifflin Harcourt Achieve Inc.

Carter, J., Cuevas, G., Day, R., Malloy, C., Kersaint, G., Luchin, B., McClain, K., Molix-Bailey, M., Price, J., Reynosa, R., Silbey, R., Vielhaber, M., & Willard, T. (2012). *Tennessee math connections course 3*. The McGraw-Hill Companies, Inc.

Hapner, K. (2018). *Fencing in Spaces: Constant Perimeter, Changing Area*  
<https://phet.colorado.edu/en/activities/4838>

Phet Interactive Simulation (2025). University of Colorado. Area Builder  
<https://phet.colorado.edu/en/simulations/area-builder>

Jeffrey, J(2014) Free House Designs. *The Wren - 2 bedroom house 6x9m*  
<https://freehousedesigns.wordpress.com/2014/09/13/the-wren-2-bedroom-house-6x9m>

Raveendran, B. and Gokulnath D. (2025) Byju's. Volume and Capacity  
<https://byjus.com/maths/volume-and-capacity>

Arc (2005) State of Victoria. Connecting volume and capacity  
<https://arc.educationapps.vic.gov.au/learning/sites/mcc/VCMMG225/Connecting-volume-and-capacity>

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|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>SmileRoad. "LVII P1 Volume and Capacity in Cost Estimation   GR11 SHS   Gen Math   Q3 MG" Youtube, 27 Sept. 2025, <a href="https://www.youtube.com/watch?v=Yh6IZRFQ8Hg">https://www.youtube.com/watch?v=Yh6IZRFQ8Hg</a></p> <p>Century Tech. "Solving Problems with Volume and Capacity   Mathematics." Youtube, 6 Nov. 2023, <a href="https://www.youtube.com/watch?v=9GnHJ0w41Jo">https://www.youtube.com/watch?v=9GnHJ0w41Jo</a></p> <p>Finding the Perimeter and Area of a Room [Video]. (n.d.). YouTube. <a href="https://www.youtube.com/watch?v=2tlar274qBM">https://www.youtube.com/watch?v=2tlar274qBM</a> YouTube</p> <p><b>Materials</b></p> <ul style="list-style-type: none"> <li>• Calculators</li> <li>• Visual aids/PowerPoint presentation</li> <li>• Ruler</li> <li>• Tape measure</li> <li>• Worksheets</li> <li>• Task Cards</li> <li>• Pencil</li> <li>• Graphing/Grid Paper</li> <li>• Eraser</li> <li>• Manila Paper/Cartolina</li> </ul> |                                                                                                                                                                                                                                                                                                                 |
| <b>III. CONTENT</b>                  | <p align="center"><b>Lesson 8.1: Estimating and Measuring Lengths and Distances</b></p> <p align="center"><i>(Recommended Time Allotment: 2 hours)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                 |
| <b>IV. LESSON OBJECTIVES</b>         | <p>At the end of the lesson, learners should be able to:</p> <ol style="list-style-type: none"> <li>1. identify and use appropriate units and tools for measuring length and distance;</li> <li>2. estimate the length or distance of given objects or spaces;</li> <li>3. accurately measure the length or distance of given objects or spaces using standard tools; and</li> <li>4. apply estimation and measurement skills in real-life tasks (e.g., determining fabric needed for clothing, measuring a room).</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                 |
| <b>V. PROCEDURES</b>                 | <p align="center"><b>ACTIVITIES</b></p> <p align="center">(Instructions for Learners)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p align="center"><b>ANNOTATION</b></p> <p align="center">(Instructions for Teachers)</p>                                                                                                                                                                                                                       |
| <b>A. Activating Prior Knowledge</b> | <b>A.1. Activating Prior Knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                 |
|                                      | <p align="center"><b>Activity A.1.1: Guess It Right!</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. The class will be divided into five groups.</li> <li>2. Five objects will be shown by the teacher.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p align="center"><b>Activity A.1.1: Guess It Right</b></p> <p>This activity is designed to strengthen learners' skills in estimation and measurement by engaging them in a playful yet educational guessing game. By observing common objects without the aid of measuring tools, learners practice making</p> |

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|  | <p>3. For each object, write down your best guess of its length, height, or width in centimeters or meters.</p> <ul style="list-style-type: none"> <li>• Length of the chalk</li> <li>• Width of the notebook</li> <li>• Height of the student</li> <li>• Width of the box</li> <li>• Length of the cellphone</li> </ul> <p>4. Do not use any measuring tools, just rely on your observation and estimation skills.</p> <p>5. After all guesses are written, the actual measurements will be revealed.</p> <p>6. Compare your guesses with the actual measurements and find out how close you were.</p> <p>7. Reflect on your activity by answering the questions below.</p> <p><b>Processing Questions:</b></p> <ol style="list-style-type: none"> <li>1. Which of your group's guesses was closest to the actual measurement?</li> <li>2. What strategy did you use to make your estimates?</li> <li>3. Which object was the most difficult to estimate accurately? Why?</li> <li>4. How did your group decide on the final estimate for each object?</li> <li>5. What did you learn about the importance of estimation in measurement?</li> </ol> | <p>reasonable approximations based on their visual perception and prior knowledge of object sizes.</p> <p>The purpose of this activity is to activate learners' prior knowledge and understanding of measurement by engaging them in estimating the lengths, widths, and heights of familiar objects. Through this guessing activity, learners will:</p> <ul style="list-style-type: none"> <li>• Practice making reasonable estimations without using measuring tools.</li> <li>• Develop awareness of measurement units (centimeters and meters).</li> <li>• Recognize the importance of accuracy and observation in estimation.</li> <li>• Reflect on the strategies used to make accurate guesses.</li> <li>• Build readiness for upcoming lessons on precise measurement and the use of measuring instruments.</li> </ul> <p>To facilitate this activity, the teacher may apply one of the following strategies below:</p> <p>Strategy 1: <b>Game-Based Learning</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Prepare the following: <ul style="list-style-type: none"> <li>• chalk</li> <li>• notebook</li> <li>• student</li> <li>• box</li> <li>• cellphone</li> </ul> </li> <li>2. Ensure a measuring tape, ruler, or meter stick is available for verifying the actual measurements.</li> <li>3. Prepare the TV screen or projector to display the actual measurements after the guessing round.</li> <li>4. Divide the class into five groups with equal members.</li> <li>5. Provide each group with paper and pen to record their guesses.</li> <li>6. Announce to the class that they will guess the measurement (length, height, or width) of different objects without using any tools.</li> </ol> |
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|  |                                                                                                                                                                                                                                                                                                                                                                                                                                     | <div>7. Allow groups 1–2 minutes to discuss and write down their best guess in centimeters or meters.</div> <div>8. After all guesses are recorded, reveal the actual measurement of each object on the TV screen.</div> <div>9. Let learners compare their guesses with the actual values.</div> <div>10. Encourage them to reflect by answering the processing questions.</div> <div>Strategy 2: <b>Concrete Experience Strategy</b></div> <div>Instructions:</div> <div><div>1. Divide the class into five groups</div><div>2. Present the following:<div><div>• chalk</div><div>• notebook</div><div>• student</div><div>• box</div><div>• cellphone</div></div></div><div>3. Instruct each group to observe carefully and write their best guess of each object’s measurement (length, width, or height) in centimeters or meters.</div><div>4. Remind them: No measuring tools allowed—only observation and estimation.</div><div>5. After all guesses are recorded, reveal the actual measurements.</div><div>6. Ask each group to compare their estimates with the real measurements.</div><div>7. Facilitate a short sharing session through processing questions.</div></div> |
|  | <b>A.2. Establishing the Purpose of the Lesson</b>                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  | <div>Activity A.2: Designer's Dilemma</div> <div>Instructions</div> <div>1. Observe the Scenario: Look at the picture showing a real-life situation where correct measurement is essential.<div><div>• An interior designer fitting a sofa into a small living room.</div><div>• A tailor measures a person for a custom-fit suit.</div><div>• A construction worker figuring out the flooring needed for a room.</div></div></div> | <div>Activity A.2.1: Designer's Dilemma</div> <div>The Designer’s Dilemma Activity engages learners in real-life scenarios where accurate measurement is essential. By observing practical situations (e.g., tailoring, construction, interior design, packaging), learners recognize the importance of precision and estimation. The dilemma prompts critical thinking about possible consequences of incorrect measurements, encouraging learners to connect</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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|  | <ul style="list-style-type: none"><li>• A product packager designing a box for a new item.</li></ul> <p>2. Think About the Dilemma: Imagine what problems could happen if the measurements in the scenario are wrong.</p> <ul style="list-style-type: none"><li>• What would happen if the sofa was an inch too long?</li><li>• What if the tailor used the wrong body measurements?</li><li>• What if the worker ordered the wrong amount of wood?</li></ul> <p>3. Discuss and Share: Talk with your group or classmates about the possible consequences. Be ready to share your ideas with the class.</p> | <p>mathematical concepts to real-world applications. Through group discussion and sharing, the activity also develops communication, collaboration, and problem-solving skills.</p> <p>The purpose of this activity is to help learners recognize the importance of accurate measurement in real-life situations. By analyzing different scenarios where measurement plays a key role, learners will:</p> <ul style="list-style-type: none"><li>• Understand how precise measurement affects the success of everyday tasks and professions.</li><li>• Identify possible problems that arise from incorrect measurements.</li><li>• Develop critical thinking skills by connecting mathematical concepts to real-world contexts.</li><li>• Appreciate the value of measurement accuracy in design, construction, tailoring, and product creation.</li><li>• Prepare for deeper learning about proper measuring techniques and tools.</li></ul> <p>To facilitate this activity, the teacher may apply one of the following strategies below:</p> <p>Strategy 1: <b>Small group discussions</b></p> <p><b>Instructions</b></p> <ol style="list-style-type: none"><li>1. Present the Scenario: Show the class a picture or a video of a common scenario that requires precise measurement.<ul style="list-style-type: none"><li>• An interior designer trying to fit a new sofa into a small living room.</li><li>• A tailor measures a person to create a custom-fit suit.</li><li>• A construction worker calculating the amount of flooring needed for a room.</li><li>• A product packager designing a box for a new item.</li></ul></li><li>2. Pose the "Dilemma": Ask the learners to think about the problems that might arise if the measurements in that scenario were wrong. Use guiding questions to spark discussion:</li></ol> |
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|                                     |                                                                                                                                                                                                  | <ul style="list-style-type: none"><li>• "What would happen if the sofa was an inch too long?"</li><li>• "What if the tailor cut the fabric based on a wrong measurement?"</li><li>• "What if the construction worker ordered the wrong amount of wood?"</li></ul> <p>Strategy2: <b>Think-Pair-Share</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"><li>1. Set the Context (Think Phase)<ul style="list-style-type: none"><li>• Show the class a picture or short video of one of the real-life scenarios (e.g., an interior designer fitting a sofa in a small living room).</li><li>• Ask learners to silently think about the situation: "What problems might happen if the measurements are wrong?"</li><li>• Give them 2–3 minutes to jot down their individual thoughts in their notebook.</li></ul></li><li>2. Pair Discussion (Pair Phase)<ul style="list-style-type: none"><li>• Instruct learners to turn to a seatmate and share their ideas.</li><li>• Encourage them to compare answers: "Did you think of the same problem? What different ideas did you notice?"</li><li>• Allow 3–4 minutes for this exchange.</li></ul></li><li>3. Group/Class Sharing (Share Phase)<ul style="list-style-type: none"><li>• Call on pairs to share their reflections with the larger group or class.</li></ul></li></ol> <p><b>Note:</b><br/>Facilitate the activity to connect the lesson into a real-world scenario and present the Performance Task.</p> |
| <b>B. Instituting New Knowledge</b> | <b>B.1. Presenting Examples</b>                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                     | <p><b>Activity B.1:<br/>Measuring Station Tour Activity</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"><li>1. Look at each object or space listed in the table.</li></ol> | <p><b>Activity B.1:<br/>Measuring Station Tour Activity</b></p> <p>This activity is designed to bridge the gap between estimation skills and accurate measurement practices by letting learners move through a set of stations with familiar classroom objects</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

2. Estimate its length, width, or height (choose the most reasonable dimension to measure) without using any measuring tool. Write your estimate in the Estimated Measurement column.
3. Use the appropriate measuring tool (ruler, meter stick, or tape measure) to find the Actual Measurement of the same object or space. Record it in the Actual Measurement column.
4. Find the Difference. Write the result in the last column.

| Object/<br>Space   | Estimated<br>Measurement<br>(cm) | Actual<br>Measurement<br>(cm) | Difference |
|--------------------|----------------------------------|-------------------------------|------------|
| Teacher's<br>Table |                                  |                               |            |
| Blackboard         |                                  |                               |            |
| Door               |                                  |                               |            |
| Classroom<br>Floor |                                  |                               |            |
| Window<br>Jamb     |                                  |                               |            |

**Processing Questions:**

1. Which of your estimates came closest to the actual measurement? Why do you think you were able to estimate it accurately?
2. Which object or space was most difficult to estimate? What made it challenging?
3. What did you learn about the use of measurement tools (ruler, tape measure, meter stick)? Which tool was most useful for this activity, and why?

and spaces. Learners begin by making educated guesses about dimensions, then verify them using appropriate measuring tools, and finally calculate the difference between their estimates and the actual results.

The purpose of this activity is to help learners understand the relationship between estimation and actual measurement through hands-on experience. By observing, estimating, and measuring real objects around them, learners will:

- Apply estimation skills to everyday items and spaces.
- Practice the correct use of measurement tools such as rulers, meter sticks, and tape measures.
- Compare estimated and actual measurements to recognize the importance of accuracy.
- Develop critical thinking and problem-solving skills in identifying sources of measurement differences.

To facilitate this activity, the teacher may apply one of the following strategies below.

**Strategy1: Rotational Learning**

**Instructions:**

1. Arrange different measuring tools (ruler, meter stick, tape measure) for the activity.
2. Identify and label the measurement stations: Teacher's Table, Blackboard, Door, Classroom Floor, and Window Jamb.
3. Prepare the activity table for learners to fill in.
4. Tell learners: "Today you will go on a Measuring Station Tour. You will estimate, measure, and compare actual sizes of familiar classroom objects and spaces."
5. Remind them that estimation comes first, and they should not use measuring tools until instructed.
6. Ask them to write down their best estimates (in centimeters) for the length, width, or height of each object/space in the Estimated Measurement column.
7. Let learners measure the same objects accurately and record the Actual Measurement in the table.

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|  |  | <p>8. Ask them to compute the Difference between their estimate and the actual measurement.</p> <p>9. Facilitate processing questions.</p> <p>Strategy2:</p> <p><b>Abstract Conceptualization and Active Experimentation</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>Set up five measuring stations around the classroom, each assigned with one object/space:             <ol style="list-style-type: none"> <li>teacher's Table</li> <li>Blackboard</li> <li>Door</li> <li>Classroom Floor</li> <li>Window Jamb</li> </ol> </li> <li>Place the required measuring tool (ruler, tape measure, or meter stick) at each station.</li> <li>Prepare a data recording sheet with columns for Estimated Measurement, Actual Measurement, and Difference.</li> <li>Tell learners: "You will go on a Measuring Station Tour. At each station, you will first estimate, then measure, and finally record your results. Rotate as a group so that everyone has a chance to participate."</li> <li>Review rules: estimate first (no tools), measure second (with tools), then compute the difference.</li> <li>Divide the class into five groups of equal members.</li> <li>Assign each group a starting station.</li> <li>At each station, groups:             <ol style="list-style-type: none"> <li>Estimate the dimension (length /width /height).</li> <li>Record it in the "Estimated Measurement" column.</li> <li>Use the measuring tool provided to get the Actual Measurement.</li> <li>Compute and record the Difference.</li> <li>After 5 minutes (or teacher's set time), give a signal (e.g., bell/hand clap) for groups to rotate clockwise to the next station.</li> </ol> </li> </ol> |
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| <b>B.2. Discussing the Concept</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                    | <p style="text-align: center;"><b>Activity B.2:</b><br/><b>Lesson Discussion on Estimate and Measure of Lengths and Distances</b></p> <p><b>Instructions</b><br/>Listen attentively and participate actively in the discussion</p> <p><b>Estimating Length and Distance</b></p> <p><b>Estimation</b> is the process of making an approximate calculation or judgment of a value, number, or quantity when an exact answer is not necessary or practical.</p> <p>We can estimate lengths and distances using approximate measurements. For example, one meter is about the distance from your shoulder to your fingertips when your arm is stretched out. It is also roughly the length of one big step or jump.</p> <p><b>Measurement</b> is a fundamental concept in mathematics that deals with determining the size, length, or quantity of an object or space.</p> <p>To measure accurately, we use measuring instruments. Some examples are given below.</p> <p>A <b>ruler</b> usually shows measurements in centimeters and millimeters. Most rulers are either 15 cm or 30 cm long. They are commonly used to measure small objects, such as a tin can, or the length of a piece of paper.</p> <p>A <b>measuring tape</b> is marked with centimeters and meters. It is useful for measuring lengths of fabric, as well as large household objects such as furniture and rooms.</p> <p>A <b>trundle wheel</b> has a circumference of 1 meter. As it is rolled along the floor, it makes a 'click' sound with every full rotation, representing 1 meter measured. Trundle wheels are useful for</p> | <p style="text-align: center;"><b>Activity B.2:</b><br/><b>Lesson Discussion on Estimate and Measure of Lengths and Distances</b></p> <p>This lesson helps learners understand the concepts of estimation and measurement and how to apply them in real-life situations. By exploring different measuring tools and solving practical problems, learners learn to estimate first, measure accurately, and calculate costs or quantities. The activity strengthens their problem-solving, reasoning, and unit-conversion skills while emphasizing the importance of accuracy and practical decision-making in everyday tasks.</p> <p>The purpose of this activity is to deepen learners' understanding of estimation and measurement by exploring real-life examples and mathematical applications. Through guided discussion and problem-solving, learners will:</p> <ul style="list-style-type: none"> <li>• Differentiate between estimation and actual measurement.</li> <li>• Recognize the importance of using appropriate tools (ruler, measuring tape, trundle wheel, odometer) for different measurement situations.</li> <li>• Apply estimation strategies using body references and familiar objects when standard tools are unavailable.</li> <li>• Perform basic calculations involving lengths and distances to connect estimation and accurate measurement.</li> <li>• Appreciate how estimation and measurement are essential skills in everyday activities such as design, construction, and travel.</li> </ul> <p>The teacher may provide more examples if necessary.</p> <p>To facilitate this lesson, the teacher may apply one of the following strategies below:</p> |

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|  | <p>measuring longer distances, such as the length of a classroom, corridor, or field.</p> <p>An <b>odometer</b> (pronounced oh-dom-uh-ter) is an instrument in cars that measures the distance travelled. The reading increases by one unit for every kilometer driven. In the example shown, the odometer indicates that the car has travelled a total of 100,000 kilometers.</p> <p><b>For example:</b><br/>The class is helping their teacher arrange the reading corner in the classroom. They need to measure different objects, but they don't have a ruler or measuring tape at first. Instead, they use body parts and familiar objects to estimate the measurements.</p> <ol style="list-style-type: none"> <li>1. The length of the teacher's desk is about 5 hand spans. If 1 hand span is about 20 cm, how long is the desk in centimeters?</li> <li>2. The height of the bookshelf is about 8-foot lengths. If 1 foot length is about 25 cm, what is the actual height of the bookshelf?</li> <li>3. The width of the doorway is about the same as 2 arm spans of Miguel. If 1 arm span is approximately 1 meter, what is the width of the doorway in meters?</li> <li>4. The class rug is estimated to be 12 books long. If each book is 30 cm long, what is the actual length of the rug?</li> <li>5. The bulletin board is about 3 notebooks wide. If each notebook is 15 cm wide, what is the actual width of the bulletin board?</li> </ol> <p><b>Expected Answers:</b></p> <ol style="list-style-type: none"> <li>1. <math>5 \times 20 \text{ cm} = 100 \text{ cm}</math> (1 meter)</li> <li>2. <math>8 \times 25 \text{ cm} = 200 \text{ cm}</math> (2 meters)</li> <li>3. <math>2 \times 1 \text{ m} = 2 \text{ m}</math></li> <li>4. <math>12 \times 30 \text{ cm} = 360 \text{ cm}</math> (3.6 meters)</li> <li>5. <math>3 \times 15 \text{ cm} = 45 \text{ cm}</math></li> </ol> | <p>Strategy 1: <b>Interactive Demonstration</b><br/><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Begin the lesson by engaging learners in a discussion about estimation and measurement, allowing them to share prior experiences or strategies they use to estimate lengths and distances.</li> <li>2. Demonstrate how to estimate using body parts or familiar objects, then show how to measure accurately using standard tools such as rulers, measuring tapes, or trundle wheels.</li> <li>3. Encourage learners to actively participate by estimating and measuring different classroom objects during the demonstration.</li> <li>4. Facilitate guided practice where learners compare their estimates with actual measurements and discuss why differences occur.</li> <li>5. Reinforce learning by summarizing key points and helping learners connect the concept of estimation and measurement to real-life applications.</li> </ol> <p>Strategy 2: <b>Guided Problem-Solving</b><br/><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. <i>Set the Context</i> <ol style="list-style-type: none"> <li>a. Begin the lesson by saying:<br/><i>"Today, we will learn how to estimate and measure lengths and distances accurately and explore the tools that can help us."</i></li> <li>b. Briefly explain why estimation and measurement are important in real-life situations (e.g., buying fabric, arranging furniture, or measuring a room).</li> </ol> </li> <li>2. <i>Introduce Estimation</i> <ol style="list-style-type: none"> <li>a. Demonstrate how to estimate a length using a practical example:               <ol style="list-style-type: none"> <li>i. Stretch your arm from shoulder to fingertips to show roughly 1 meter.</li> <li>ii. Take a "big step" to estimate distances.</li> </ol> </li> </ol> </li> </ol> |
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|  | <p>Problem: A trundle wheel is rolled along the length of a classroom corridor, and it clicks 12 times.</p> <ol style="list-style-type: none"> <li>1. How long is the corridor in meters?</li> <li>2. If the floor tiles used are 60 cm long each, how many tiles are needed to cover the corridor lengthwise?</li> </ol> <p>Solution</p> <p>Given:</p> <ul style="list-style-type: none"> <li>• 1 click of trundle wheel = 1 meter</li> <li>• Number of clicks = 12</li> <li>• Length of one floor tile = 60 cm = 0.6 m</li> </ul> <ol style="list-style-type: none"> <li>1. To find the length of the corridor, note that<br/> Length of corridor = Number of clicks × Length per click<br/> <math>= 12 \times 1\text{m} = 12\text{m}</math></li> </ol> <p>Thus, the corridor is 12metres long.</p> <ol style="list-style-type: none"> <li>2. To find how many floor tiles needed,<br/> Note that each tile is 0.6m long, then the corridor is 12m long. We have<br/> <math display="block">\begin{aligned} \text{Number of tiles} &amp;= \frac{\text{Length of corridor}}{\text{Length of one tile}} \\ &amp;= \frac{12\text{m}}{0.6\text{m}} \\ &amp;= 20 \end{aligned}</math></li> </ol> <p>Thus, 20 tiles are needed to cover the corridor lengthwise.</p> <p><b>Problem:</b> A family's car odometer shows 57,845 km before a trip. After the trip, it reads 58,234 km.</p> <ol style="list-style-type: none"> <li>1. How far did the family travel?</li> <li>2. If the car consumes 1 liter of gasoline for every 12 km, how many liters were consumed on the trip?</li> </ol> <p><b>Given:</b></p> <ul style="list-style-type: none"> <li>• Odometer reading before trip: 57,845 km</li> <li>• Odometer reading after trip: 58,234 km</li> <li>• Fuel consumption rate: 1 liter per 12 km</li> </ul> | <ol style="list-style-type: none"> <li>b. Ask learners to guess the length of a classroom object using this method.</li> </ol> <p>3. <i>Demonstrate Measurement Tools</i></p> <ol style="list-style-type: none"> <li>a. Show each tool, explain its purpose, and demonstrate how it works: <ol style="list-style-type: none"> <li>i. Ruler – for small objects (e.g., paper, tin).</li> <li>ii. Measuring tape – for medium distances (e.g., fabric, furniture).</li> <li>iii. Trundle wheel – for longer distances (e.g., classroom, corridor).</li> <li>iv. Odometer – for tracking distance traveled by vehicles.</li> </ol> </li> <li>b. Emphasize units of measurement (cm, m, km) and when each tool is most appropriate.</li> </ol> <p>4. <i>Guided Practice</i></p> <ol style="list-style-type: none"> <li>a. Give learners objects to measure individually or in pairs.</li> <li>b. Ask them to estimate the measurement first, measure accurately using the appropriate tool and compare their estimates with the actual measurement.</li> </ol> <p>5. <i>Connect to Real-Life Examples</i></p> <ol style="list-style-type: none"> <li>a. Present a practical example (e.g., calculating fabric for a dress or ribbon for decorating a room).</li> <li>b. Demonstrate step-by-step how to calculate total material needed and cost.</li> <li>c. Ask learners to solve a similar example with teacher's guidance.</li> </ol> |
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**Solutions**

1. To find the distance travelled, we have  
Distance travelled = Final reading – Initial reading  
= 58,234 km - 57,845 km  
= 389 km

Thus, the family travelled 389 kilometres.

2. To find the amount of gasoline consumed, we have

$$\begin{aligned}\text{Fuel consumed} &= \frac{\text{Distance travelled}}{\text{Distance per litre}} \\ &= 389\text{km} \cdot \frac{1\text{L}}{12\text{km}} \\ &= 32.4166 \text{ or } 32.42\text{L}\end{aligned}$$

Thus, the car consumed approximately 32.42 liters of gasoline.

**B.3. Developing Mastery**

**Activity B.3:  
Practicing the Skills on Estimating and Measuring  
Lengths and Distances**

**Instructions:**

Read and follow the instructions carefully. Work collaboratively with your group in completing each part of the activity. Use appropriate tools and units of measurement when required. Record all your observations and answers neatly in the activity sheet provided. Be sure to compare your estimates with the actual measurements and reflect on your findings through the given processing questions.

*Part A: Estimation*

1. Each group shall select five (5) objects within the classroom or nearby areas to be estimated.
2. Using only familiar items inside your bag (such as a ballpen, notebook, etc.) estimate the length or distance of each chosen object.
3. Record all estimated lengths or distances in the activity sheet provided.

| Object | Estimated Measurement | Unit Used (Familiar Object) | Remarks |
|--------|-----------------------|-----------------------------|---------|
|--------|-----------------------|-----------------------------|---------|

**Activity B.3:  
Practicing the Skills on Estimating and Measuring  
Lengths and Distances**

This activity provides learners with opportunities to apply measurement concepts and problem-solving skills to practical, real-world scenarios. By working on word problems involving ribbons, cloth, corridor length, and travel distance, learners reinforce their understanding of estimation, unit conversion, and arithmetic operations in measurement contexts.

This activity is to help learners develop mastery in estimating and measuring lengths or distances through hands-on and collaborative practice. Learners will strengthen their understanding of the relationship between estimation and actual measurement by:

- Applying learned concepts of estimation using familiar objects as nonstandard units.
- Accurately measuring objects with standard measuring tools such as rulers, measuring tapes, or trundle wheels.



|    |  |  |  |
|----|--|--|--|
| 1. |  |  |  |
| 2. |  |  |  |
| 3. |  |  |  |
| 4. |  |  |  |
| 5. |  |  |  |

### *Part B: Measurement*

1. Using appropriate measuring instruments such as a ruler, measuring tape, or trundle wheel, measure the actual length or distance of each object selected.
2. Record the actual measurements beside the corresponding estimates.

| Object | Estimated Measurement | Actual Measurement | Measuring Tool Used |
|--------|-----------------------|--------------------|---------------------|
| 1.     |                       |                    |                     |
| 2.     |                       |                    |                     |
| 3.     |                       |                    |                     |
| 4.     |                       |                    |                     |
| 5.     |                       |                    |                     |

### *Part C: Comparison and Reflection*

1. Compute the difference between the estimated and actual measurements.
2. Reflect on your results by answering the processing questions below.

| Object | Estimated Measurement | Actual Measurement | Difference | More Accurate Estimation (Yes/No) |
|--------|-----------------------|--------------------|------------|-----------------------------------|
| 1.     |                       |                    |            |                                   |
| 2.     |                       |                    |            |                                   |
| 3.     |                       |                    |            |                                   |
| 4.     |                       |                    |            |                                   |
| 5.     |                       |                    |            |                                   |

### **Processing Questions:**

1. Which of your estimates were closest to the actual measurements?

- Comparing estimated and actual measurements to evaluate accuracy and precision.
- Reflecting on the importance of using standard units and appropriate measuring tools.
- Working cooperatively to enhance critical thinking, communication, and problem-solving skills in practical measurement tasks

To facilitate this activity, the teacher may apply one of the following strategies below:

### **Strategy1: Concrete Experience Strategy**

#### **Instructions:**

1. Allow learners to engage directly in the activity by selecting five (5) real objects within or near the classroom to estimate and measure.
2. Encourage them to use familiar items (e.g., pencil, notebook, ID, eraser) for estimation before applying standard measuring tools.
3. Let learners record their observations and data in the activity sheet.
4. Emphasize active participation, collaboration, and real-life application.

### **Strategy2: Guided Discovery Learning Strategy**

#### **Instructions:**

1. Organize learners into small groups to promote collaboration, sharing of ideas, and cooperative problem-solving.
2. Guide the learners as they perform the activity by providing prompts or questions that help them think critically, make predictions, and reflect on their observations.
3. Encourage learners to use familiar objects for estimating measurements before using standard tools to verify their results.



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|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | <p>2. What familiar object gave you the most accurate estimates? Why?</p> <p>3. What did you learn about the importance of using standard units of measurement?</p> <p>4. How can estimation be useful in daily life situations?</p> <p>5. How did teamwork help in completing this activity?</p>                                                                                                                                                                                                                                                                                                                         | <p>4. Encourage learners to use familiar objects for estimating measurements before using standard tools to verify their results.</p> <p>5. Facilitate discussion after the activity by letting learners share their findings, compare their estimates and actual measurements, and identify factors affecting accuracy.</p> <p>6. Help learners construct understanding by connecting their discoveries to mathematical concepts such as accuracy, precision, and the importance of standard units.</p> <p>7. Encourage application by relating the lesson to practical and real-world situations where estimation and measurement are needed.</p> <p>8. Assess learning based on participation, reasoning, accuracy of results, and reflection on the discovery process.</p> <p><b>Reminder</b><br/>Act as a facilitator, guiding learners toward constructing knowledge through exploration, questioning, and reflection rather than direct instruction.</p> |
| <b>C.<br/>Demonstrating<br/>Knowledge and<br/>Skills</b> | <b>C.1. Finding Practical Application</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                          | <p><b>Activity C.1: Estimating Lengths Around You</b></p> <p><b>Instructions:</b><br/>Work as a group. Perform the required tasks by making estimates first, then measuring accurately.</p> <p><b>Task 1: Measuring the Whiteboard</b></p> <ul style="list-style-type: none"> <li>You are asked to estimate the length of your Use your arm length or handspan to measure how wide the whiteboard is.</li> <li>Record how many arm lengths or handspans it takes to cover the whole whiteboard.</li> <li>After estimating, use a ruler or meter stick to find the actual measurement.</li> </ul> <p><b>Questions:</b></p> | <p><b>Activity C.1: Estimating Lengths Around You</b></p> <p>This activity helps learners connect estimation and measurement concepts to real-life situations through hands-on tasks in familiar school settings. It enhances their accuracy, critical thinking, and understanding of measurement tools while promoting collaboration, reflection, and appreciation of mathematics in solving everyday problems.</p> <p>This activity is to help learners connect their understanding of estimation and measurement to real-life situations by applying these skills in familiar environments. Through hands-on tasks, learners will:</p> <ul style="list-style-type: none"> <li>Practice estimating and measuring objects and distances using both nonstandard and standard tools.</li> </ul>                                                                                                                                                                  |

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|          | <ul style="list-style-type: none"><li>• Was your estimate close to the actual length?</li><li>• What might have caused the difference?</li></ul> <p><b>Task 2: Measuring the Distance to the Flagpole</b></p> <ul style="list-style-type: none"><li>• Your group will estimate the distance from your classroom door to the flagpole in the school ground.</li><li>• Use footsteps to estimate how far the flagpole is from your classroom.</li><li>• Count your steps as you walk naturally toward the flagpole.</li><li>• Compare your estimated distance to the actual measurement using a tape measure.</li></ul> <p><b>Question:</b></p> <ul style="list-style-type: none"><li>• How accurate was your estimation using footsteps?</li></ul> <p><b>Task 3: Measuring the classroom door</b></p> <ul style="list-style-type: none"><li>• You are asked to estimate the length of a bench found in the classroom door.</li><li>• Use a handspan or a notebook as your measuring tool.</li><li>• Count how many handspans or notebooks long the bench is.</li><li>• Compare your estimated distance to the actual measurement using a meterstick.</li></ul> <p><b>Question:</b></p> <ul style="list-style-type: none"><li>• Which tool (handspan or notebook) gave a more accurate result? Why?</li></ul> <p><b>Rubrics:</b></p> <table><tr><td>Criteria</td><td>4 –<br/>Excellent</td><td>3 –<br/>Proficient</td><td>2 –<br/>Developing</td><td>1 –<br/>Beginning</td></tr></table> |                   |                   |                  |  | Criteria | 4 –<br>Excellent | 3 –<br>Proficient | 2 –<br>Developing | 1 –<br>Beginning | <ul style="list-style-type: none"><li>• Develop accuracy and awareness of factors that affect measurement results.</li><li>• Recognize the importance of estimation and measurement in daily activities and problem-solving.</li><li>• Strengthen teamwork, observation, and critical thinking skills through collaborative exploration and reflection.</li></ul> <p>To facilitate this activity, follow the instructions below.</p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"><li>1. Group the learners into five (5).</li><li>2. Present the three scenarios (whiteboard, flagpole, classroom door/bench).</li><li>3. Let learners first estimate the needed measurements.</li><li>4. Provide the actual measuring tools (tape measure, ruler, meter stick).</li><li>5. Have learners measure accurately, calculate totals, and compare results with their estimates.</li><li>6. Guide them to generalize: What do we learn about the role of estimation vs. accurate measurement?</li><li>7. Encourage learners to apply their learning by discussing other real-life situations where measurement is important.</li><li>8. Let pairs or groups present their results and reflections.</li></ol> |  |  |  |  |
| Criteria | 4 –<br>Excellent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3 –<br>Proficient | 2 –<br>Developing | 1 –<br>Beginning |  |          |                  |                   |                   |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |  |  |

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|  | <b>Accuracy of Estimation and Measurement</b>  | Estimates are highly accurate and very close to the actual measurements (difference of less than 5%). Shows clear understanding of measurement concepts. | Estimates are mostly accurate and close to actual measurements (difference of 5–10%). Demonstrates good understanding of measurement concepts. | Estimates show partial accuracy but significant differences from actual measurements (difference of 10–20%). Shows developing understanding. | Estimates are far from actual measurements (difference greater than 20%). Shows minimal understanding of measurement concepts. |  |  |
|  | <b>Use of Appropriate Tools and Techniques</b> | Consistently uses suitable tools (arm length, footsteps, notebook, etc.) effectively and records data neatly and correctly.                              | Uses appropriate tools correctly most of the time and records measurements with minor errors.                                                  | Uses tools inconsistently or with some errors in measurement or recording.                                                                   | Uses inappropriate tools or fails to record measurements accurately.                                                           |  |  |
|  | <b>Group Collaboration and Participation</b>   | All members actively participate, share ideas, and cooperate effectively throughout the task.                                                            | Most members participate and cooperate with minimal prompting.                                                                                 | Some members participate, but others are passive or less involved.                                                                           | Little to no collaboration; task is completed by one or two members only.                                                      |  |  |
|  | <b>Data Recording and Presentation</b>         | Data is complete, well-organized, and clearly presented with accurate comparisons and neat work.                                                         | Data is mostly complete and organized with minor inaccuracies or untidy presentation.                                                          | Data is incomplete or disorganized with noticeable inaccuracies.                                                                             | Data is missing, unclear, or poorly presented.                                                                                 |  |  |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <table><tr><td><b>Reflection and Analysis</b></td><td>Responses to processing questions are insightful, clearly explaining accuracy, differences, and real-life application of estimation.</td><td>Responses show clear understanding with some reflection on accuracy and differences.</td><td>Responses show limited reflection or partial understanding of results.</td><td>Responses are minimal, unclear, or missing.</td></tr></table> | <b>Reflection and Analysis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Responses to processing questions are insightful, clearly explaining accuracy, differences, and real-life application of estimation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Responses show clear understanding with some reflection on accuracy and differences. | Responses show limited reflection or partial understanding of results.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Responses are minimal, unclear, or missing. |  |
| <b>Reflection and Analysis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Responses to processing questions are insightful, clearly explaining accuracy, differences, and real-life application of estimation.                                                                                                                                                                                                                                                                                                         | Responses show clear understanding with some reflection on accuracy and differences.                                                                                                                                                                                                                                                                                                                                                                                                                                     | Responses show limited reflection or partial understanding of results.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Responses are minimal, unclear, or missing.                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                             |  |
| <b>Scoring Guide:</b><br>16–20 points: Excellent understanding and application<br>11–15 points: Proficient performance<br>6–10 points: Developing skills and understanding<br>1–5 points: Beginning stage; needs significant improvement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                             |  |
| <b>C.2. Making Generalization</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                             |  |
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|  |  | <p>To facilitate this activity, you may apply one of the strategies below.</p> <p><b>Strategy 1: Guided Reflection Questions</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Tell learners to pause and reflect on what we have learned about estimation and measurement.</li> <li>2. Present the guide questions and incomplete sentences on the board or projector.</li> <li>3. Answer these prompts based on your own understanding and experiences from our activities.</li> <li>4. Ask learners to work silently for a few minutes to complete each sentence.</li> <li>5. Highlight similarities and differences in responses.</li> <li>6. Work with the class to build a collective summary for each prompt.</li> <li>7. Write down key points on the board as the “class generalization.”</li> </ol> <p><b>Strategy 2: Guided Discovery Learning Strategy</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Begin by saying: “Let’s recall what we experienced in our previous activities about estimating and measuring. Think about what you did and what you discovered.”</li> <li>2. Show the incomplete sentences/questions (estimation, measurement, tools, and accuracy).</li> <li>3. Tell learners to construct your own meaning by answering these prompts using your experiences.</li> <li>4. Group learners into pairs or small groups.</li> <li>5. Ask them to discuss and complete the sentences together.</li> <li>6. Encourage them to explain their reasoning and use examples from the activities.</li> <li>7. Let groups present their completed sentences.</li> <li>8. Record their answers on the board</li> </ol> |
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|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 9. Together, synthesize the responses into a class-agreed generalization for each question.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>C.3. Evaluating Learning</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 | <p style="text-align: center;"><b>Activity C.3: Assessing Mastery</b></p> <p><b>Instructions:</b><br/>Read and answer each item.</p> <ol style="list-style-type: none"> <li>Which of the following best defines estimation?<br/>A. Converting one unit to another<br/>B. Measuring an object using a ruler<br/>C. Calculating the exact length of an object<br/>D. Making an approximate judgment of a value or quantity</li> <li>If one meter is approximately the distance from your shoulder to your fingertips, what does this show?<br/>A. The use of a trundle wheel<br/>B. A standard measuring tool<br/>C. The length of a small object<br/>D. An example of estimation using a familiar reference</li> <li>Which of the following is the best tool to measure the length of a football field?<br/>A. Ruler<br/>B. Tape measure<br/>C. Trundle wheel<br/>D. Odometer</li> <li>A ribbon costs ₱20 per meter. If you need 4.5 meters, how much will you pay?<br/>A. ₱80      B. ₱85      C. ₱90      D. ₱100</li> <li>An odometer reading increased from 25,320 km to 25,645 km after a trip. How far did the car travel?<br/>A. 225 km    B. 315 km    C. 325 km    D. 335 km</li> <li>Which of the following situations best shows when estimation is useful?<br/>A. Exact measurement is always required<br/>B. We want to save time and get a quick idea of size or distance<br/>C. We are building machines that require precision</li> </ol> | <p style="text-align: center;"><b>Activity C.3: Assessing Mastery</b></p> <p>This activity assesses whether learners have mastered key concepts of estimation, measurement, and practical applications. Through multiple-choice questions, learners demonstrate their ability to choose appropriate tools, perform calculations, and apply estimation and measurement in real-life contexts. It reinforces the connection between classroom learning and everyday tasks while providing a quick and effective way to evaluate understanding. Overall, it ensures that learners can reason, apply, and reflect on what they have learned.</p> <p>Answer Key:</p> <ol style="list-style-type: none"> <li>D</li> <li>C</li> <li>C</li> <li>B</li> <li>B</li> <li>C</li> <li>C</li> <li>D</li> <li>D</li> <li>D</li> </ol> |

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|                                   | <p>D. We cannot use measuring tools at all</p> <p>7. A dressmaker needs 2.5 meters of fabric for one dress. If she makes 3 dresses, how many meters of fabric are needed in total?<br/>A. 5.5 m    B. 6.5 m    C. 7.5 m    D. 8.5 m</p> <p>8. Why might two learners measuring the same desk using hand spans get different results?<br/>A. The desk changed size<br/>B. They used different rulers<br/>C. Their hand spans are not of equal length<br/>D. One student miscounted the desk's corners</p> <p>9. Which statement best explains the advantage of using standard measuring tools over body parts?<br/>A. They are faster to use.<br/>B. They are less expensive.<br/>C. They are smaller and lighter.<br/>D. They are more accurate and consistent.</p> <p>10. You are asked to measure the length of a school playground but do not have a measuring tape. What is the best strategy to obtain an approximate measurement?<br/>A. Guess the length by looking at it<br/>B. Wait until someone brings a ruler<br/>C. Use your notebook to measure repeatedly<br/>D. Use large steps or a trundle wheel if available to estimate the distance.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>C.4. Additional Activities</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                   | <p style="text-align: center;"><b>Activity C.4:<br/>More Real-Life Problems on Measurement</b></p> <p>Instructions:<br/>Solve the following problems. Show your complete solutions.</p> <p><b>For Remediation</b></p> <p>1. Mark estimates the width of the window, using arms, to be about 1.9 meters. At the shop, the material costs ₱120 per</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p style="text-align: center;"><b>Activity C.4:<br/>More Real-Life Problems on Measurement</b></p> <p>This activity supports both enrichment and remediation by engaging learners in real-life problem-solving that applies estimation, measurement, and computation skills. It helps struggling learners strengthen their understanding through guided practice and concrete examples, while providing advanced learners with opportunities for higher-order</p> |

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|              | <p>meter.</p> <ol style="list-style-type: none"> <li>How many meters of material should Carl buy?</li> <li>How much will the material total cost?</li> </ol> <p>2. A carpenter estimates that the length of a wooden plank is about 1.7 meters. When measured accurately, it is 1.9 meters.</p> <ol style="list-style-type: none"> <li>How many meters of wood should he pay for if wood is sold only by the meter?</li> <li>If the price is ₱120 per meter, how much will he spend?</li> </ol> <p><b>For Enrichment</b></p> <p>3. A family's car odometer reads 48,625 km before a road trip. After reaching their destination, it reads 49,040 km.</p> <ol style="list-style-type: none"> <li>How many kilometers did the family travel?</li> <li>If the car consumes 1 liter of gasoline for every 14 km, how many liters of gasoline were used for the trip?</li> <li>If gasoline costs ₱68.50 per liter, how much did the family spend on fuel for the trip?</li> </ol> | <p>thinking and real-world application. Overall, the task enhances mathematical reasoning, accuracy, and problem-solving skills through meaningful and practical contexts.</p> <p>The purpose of this activity is to enable learners to apply their understanding of estimation, measurement, and basic mathematical operations to solve real-life problems. Through these tasks, learners will:</p> <ul style="list-style-type: none"> <li>Demonstrate the ability to use estimation and precise measurement in practical situations.</li> <li>Apply problem-solving strategies involving length, distance, and cost computation.</li> <li>Strengthen their skills in converting real-world contexts into mathematical expressions.</li> <li>Recognize the importance of accuracy and estimation in everyday decision-making.</li> <li>Develop logical reasoning and analytical thinking through step-by-step problem-solving.</li> </ul> <p>For remediation, provide learners with simpler one- or two-step problems that focus on basic estimation, measurement, and computation skills. For learners who need enrichment, assign more complex, multi-step problems that require analysis, reasoning, and the extended application of estimation and measurement in practical contexts.</p> <p><b>Answer Key</b></p> <ol style="list-style-type: none"> <li> <ol style="list-style-type: none"> <li>2 meters</li> <li>₱240.00</li> </ol> </li> <li> <ol style="list-style-type: none"> <li>2 meters</li> <li>₱240.00</li> </ol> </li> <li> <ol style="list-style-type: none"> <li>415 km</li> <li>29.64 liters</li> <li>₱2,030.34</li> </ol> </li> </ol> |
| III. CONTENT | <p align="center"><b>Lesson 8.2: Calculating Area and Perimeter for Practical Purposes</b><br/>(Recommended Time: 2 hours)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



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| <b>IV.<br/>OBJECTIVES</b>                    | At the end of the lesson, the learners are able to: <ol style="list-style-type: none"> <li>1. calculate the area and perimeter of rectangular, square, triangular, circular and composite residential spaces.</li> <li>2. solve contextual problems involving room renovation and garden lay-outing.</li> <li>3. demonstrate appreciation of the importance of area and perimeter in everyday life by explaining their usefulness in home improvement and budgeting projects.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>V.<br/>PROCEDURES</b>                     | <b>ACTIVITIES</b><br>(Instructions for Learners)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>ANNOTATION</b><br>(Instructions for Teachers)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>A. Activating<br/>Prior<br/>Knowledge</b> | <b>A.1. Activating Prior Knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                              | <p style="text-align: center;"><b>Activity A.1: Everyday Objects Around Us</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Look at the following objects that are commonly seen at home or in school:             <ul style="list-style-type: none"> <li>○ Laptop screen (rectangle)</li> <li>○ Square picture frame</li> <li>○ Round plate</li> <li>○ Car Early Warning Device</li> </ul> </li> <li>2. Estimate or measure their dimensions (length, width, or diameter).</li> <li>3. Answer the following guide questions:             <ul style="list-style-type: none"> <li>○ What formula will you use to find the area of each object?</li> <li>○ What formula will you use to find the perimeter (or circumference for the plate) of each object?</li> <li>○ How are these similar to the shapes we see in houses or gardens (e.g., walls, windows, tiles, flower beds)?</li> </ul> </li> <li>4. Share your answers with a partner or group.</li> </ol> | <p style="text-align: center;"><b>Activity A.1: Everyday Objects Around Us</b></p> <p>This activity will help learners recall and connect their knowledge of area and perimeter through real-life familiar objects. It aims to connect the lesson on area and perimeter to real-life situations that are familiar to learners. By showing figures, learners are encouraged to see practical applications of mathematics in everyday life.</p> <p>Note: Teachers can make localized examples.</p> <p>To facilitate the activity, apply any of the suggested strategies below.</p> <p>Strategy 1: <b>Experiential learning + constructivism</b></p> <ol style="list-style-type: none"> <li>1. Preparation             <ul style="list-style-type: none"> <li>● Prepare real objects (a laptop, a square picture frame, and a round plate) or show clear pictures of them.</li> <li>● rulers/tape measures for learners (if measurement is allowed).</li> <li>● Distribute worksheets with guide questions.</li> </ul> </li> <li>2. Set the Context             <ul style="list-style-type: none"> <li>● Show to the learners the four objects (laptop screen, square picture frame, round plate) and car early warning device).</li> <li>● Ask: “Have you ever thought about how much space these objects cover or how long their edges are?”</li> <li>● Connect by saying: “Today we will recall how to find the area and perimeter of shapes we see every day,</li> </ul> </li> </ol> |

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|  |  | <p>and later we'll connect this to larger spaces like rooms and gardens.”</p> <p>3. Guide the Exploration</p> <ul style="list-style-type: none"> <li>• Divide the class into small groups or pairs.</li> <li>• Instruct them to estimate or measure the dimensions: <ul style="list-style-type: none"> <li>○ Laptop screen – length and width (rectangle)</li> <li>○ Picture Frame- side length (square)</li> <li>○ Plate – diameter (circle)</li> <li>○ Car Early Warning Device- base and height (triangle)</li> </ul> </li> <li>• Encourage them to sketch each object and label the dimensions on their worksheet.</li> </ul> <p>4. Prompt Critical Thinking</p> <ul style="list-style-type: none"> <li>• Ask groups to answer these guide questions:</li> <li>• What formula will you use to find the area of each object?</li> <li>• What formula will you use to find the perimeter (or circumference for the plate) of each object?</li> <li>• How are these shapes like the ones we see in houses or gardens (e.g., walls, windows, tiles, flower beds)?</li> </ul> <p>6. Facilitate Sharing</p> <ul style="list-style-type: none"> <li>• Have learners share their group's answers with the class</li> <li>• Record key points on the board (area and perimeter formulas, real life connections).</li> </ul> <p><b>Strategy 2: Gallery Walk</b></p> <p>1. Set Up the Task</p> <ol style="list-style-type: none"> <li>a. Divide the class into small groups (3-5 learners per group)</li> <li>b. Give each group a different object.</li> <li>c. Pose the problem:<br/>“Find the area (space) and perimeter (border) of this 1 object. Show your solution clearly on manila paper/cartolina.”</li> </ol> <p>2. Display Work</p> <ol style="list-style-type: none"> <li>a. Ask groups to post their finished solutions around the room (walls, desks, chalkboards).</li> <li>b. Provide markers or sticky notes for feedback.</li> </ol> |
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|                                                    |                                                                                                                                                                                                                                                                                                                                          | <ol style="list-style-type: none"><li>3. Conduct the Gallery Walk<ol style="list-style-type: none"><li>a. Instruct groups to rotate around the room, viewing the posted solutions of the groups.</li><li>b. Give prompts:<ul style="list-style-type: none"><li>• “What strategies did they use to find the area?”</li><li>• “Is their perimeter calculation correct?”</li><li>• “What can we learn or improve from this solution?”</li></ul></li></ol></li><li>4. Whole-Class Discussion<p>Lead a debrief by asking:</p><ul style="list-style-type: none"><li>• “What similarities and differences did you notice among the solutions?”</li><li>• Which solutions showed clear reasoning?</li><li>• “How does this apply to real- life situations like painting or fencing?”</li></ul></li><li>5. Summarize their responses:<p>“So, we discovered that area and perimeter are useful for real-life like painting, covering, budgeting, or measuring capacity. Today we will learn how to calculate them using mathematical formulas.”</p></li></ol> |
| <b>A.2. Establishing the Purpose of the Lesson</b> |                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                    | <p style="text-align: center;"><b>Activity A.2:</b><br/><b>Why Measures Matter: Renovation Challenge</b></p> <p><b>Part 1: Observe &amp; Compare</b></p> <p>1. Look carefully at the pictures provided.</p> <div data-bbox="443 979 1144 1414"></div> | <p style="text-align: center;"><b>Activity A.2:</b><br/><b>Why Measures Matter: Renovation Challenge</b></p> <p>This activity will able the learners to discover why knowing area and perimeter is important in real life, and how these measurements help us make smart decisions when planning renovations or designing spaces.</p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"><li>1. <i>Introduce the Activity:</i><ul style="list-style-type: none"><li>• Explain that students will explore how measurements like area and perimeter are important in real-life renovation projects.</li><li>• Emphasize that they will practice observing, estimating, and calculating measurements.</li></ul></li><li>2. <i>Guide Observation:</i><ul style="list-style-type: none"><li>• Display the provided pictures to the class.</li></ul></li></ol>                                                                                                                                                                                 |

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|  | <p>2. Complete the following:</p> <ol style="list-style-type: none"> <li>List 3 differences you notice between the images.</li> <li>Imagine you are renovating these spaces. Identify 2 items you would need to buy. For each item, write: <ul style="list-style-type: none"> <li>The item name</li> <li>Which measurement you need: area or perimeter</li> </ul> </li> </ol> <p><b>Part 2: Renovation Scenarios</b></p> <ol style="list-style-type: none"> <li>Your teacher will present real-life renovation scenarios. <ul style="list-style-type: none"> <li>You want to paint a rectangular wall in your classroom that is 4 meters high and 6 meters wide.</li> <li>Your kitchen floor measures 3 meters by 5 meters, and you need to buy tiles to cover the entire floor.</li> <li>You want to build a fence around your rectangular backyard that measures 8 meters by 12 meters.</li> <li>You have a rectangular flower bed 2 meters wide and 5 meters long, and you want to put a decorative border around it.</li> <li>A living room measures 4 meters by 7 meters, and you plan to cover the floor with a carpet.</li> </ul> </li> <li>In your group, discuss and decide: <ul style="list-style-type: none"> <li>Is this an area or perimeter problem?</li> <li>Which formula will you use?</li> </ul> </li> <li>Try to estimate the measurement before calculating. Discuss why estimation is useful in real life.</li> </ol> <p><b>Part 3: Reflection</b></p> <p>Answer the following:</p> <ol style="list-style-type: none"> <li>What helped you decide whether to use area or perimeter?</li> <li>How can knowing these measurements help you at home, at school, or in other real-life situations?</li> <li>What questions do you have that you want your teacher to answer during the lesson?</li> </ol> | <ul style="list-style-type: none"> <li>Ask students to list 3 differences they notice between the images.</li> <li>Then, have them identify 2 items they would need to buy if renovating the spaces, and specify for each whether they would need area or perimeter to determine the amount needed.</li> <li>Circulate and support students, prompting them with questions like: <ol style="list-style-type: none"> <li>“Are you covering a surface or surrounding a space?”</li> <li>“How does the size of the space affect what you need to buy?”</li> </ol> </li> </ul> <p>3. <i>Facilitate Scenario Discussion:</i></p> <ul style="list-style-type: none"> <li>Present the provided renovation scenarios to the students.</li> <li>Have students work in groups to decide whether each scenario involves calculating area or perimeter, and identify the correct formula: <math display="block">\text{Area} = \text{length} \times \text{width}</math> <math display="block">\text{Perimeter} = 2 \times (\text{length} + \text{width})</math> </li> <li>Encourage them to estimate the measurements before calculating, discussing why estimation is useful (e.g., budgeting, buying enough materials, avoiding waste).</li> <li>Ask each group to share their reasoning for at least one scenario.</li> </ul> <p>4. <i>Encourage Reflection and Connection:</i></p> <ul style="list-style-type: none"> <li>Ask students to reflect on the following questions.</li> <li>Facilitate a brief class discussion to highlight practical applications and clarify misconceptions.</li> </ul> <p>5. <i>State the Connection and Importance:</i></p> <ul style="list-style-type: none"> <li>Reinforce that this activity demonstrates the real-life application of area and perimeter.</li> <li>Explain that understanding these measurements will help them in practical situations such as painting, tiling, fencing, or furnishing spaces.</li> </ul> |
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|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Introduce the performance task that will be given at the end of the lesson and tell learners they will apply their understanding of area and perimeter to complete it successfully.</li> </ul> <p>6. Monitor and Support:</p> <ul style="list-style-type: none"> <li>Observe student engagement during observation, discussion, and calculation.</li> <li>Check for understanding of area versus perimeter and accuracy in estimation and computation.</li> <li>Provide guidance where students struggle, using real-life examples to reinforce concepts.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>B. Instituting New Knowledge</b> | <b>B.1. Presenting Examples</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                     | <p align="center"><b>Activity B.1: Measure It Right!</b></p> <p><b>Instructions:</b><br/>The problems are examples of real-life scenarios where area and perimeter apply. Study each problem and answer the questions provided.</p> <p><b>Example 1: The Classroom Bulletin Board</b><br/>The Grade 11 learners are decorating a rectangular bulletin board that measures 2 meters long and 1.5 meters wide. They plan to:</p> <ul style="list-style-type: none"> <li>Paste colored paper to cover the whole board, and</li> <li>Put a border of ribbon around the edges.</li> </ul> <p><b>Questions:</b></p> <ol style="list-style-type: none"> <li>What quantity do they need to find if they want to know how much colored paper to buy?</li> <li>What quantity do they need to find if they want to know how long the ribbon should be?</li> </ol> <p><b>Expected Answer:</b></p> <ul style="list-style-type: none"> <li>Colored paper → Area</li> <li>Ribbon → Perimeter</li> </ul> | <p align="center"><b>Activity B.1: Measure It Right!</b></p> <p>This activity will help learners recognize the importance of measurement in everyday life and relate it to practical uses such as determining area and perimeter.</p> <p>To facilitate this activity the teacher may apply one of the following strategies.</p> <p><i>Note: Teachers can make localized examples and may add examples if necessary.</i></p> <p>Strategy 1: <b>Worked Example + Thing-Aloud+ Guided Practice</b><br/><b>Instructions:</b><br/><b>Begin by saying:</b> “Today, we will learn how to calculate the area and perimeter of real spaces. Let’s start with worked examples to understand the process clearly.”<br/><b>Step 1 (Modeling):</b> Solve Example 1 on the board while thinking aloud.<br/><b>Step 2 (Guided Practice):</b> Ask learners to help solve Example 2, prompting them to suggest steps.<br/><b>Step 3 (Independent Practice):</b> Let learners solve Example 3 on their own, then compare answers.</p> |

**Computation (for checking):**

$$\text{Area} = 2 \times 1.5 = 3 \text{ m}^2$$

$$\text{Perimeter} = 2(2 + 1.5) = 7 \text{ m}$$

**Example 2: Designing a Triangular Metal Sign Plate**

The TechPro learners at Agay National High School -Senior High School are asked to make a triangular warning sign plate for the welding area. The triangular plate has a base of 40 cm and a height of 30 cm. Each side of the triangle measures 40 cm, 50 cm, and 60 cm.

Before attaching the sign to the wall, they must:

- Paint the front surface of the metal plate, and
- Weld a thin metal border around its sides for protection.

**Questions:**

1. What quantity should the learners find to know how much paint to use?
2. What quantity should they find to know how long the welded border should be?

**Expected Answers:**

Paint → Area

Welded border → Perimeter

**Computation (for checking):**

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 40 \times 30$$

$$= 600 \text{ cm}^2$$

$$\text{Perimeter} = 40 + 50 + 60 = 150 \text{ cm}$$

**Therefore:**

- The area of the triangular sign plate is  $600 \text{ cm}^2$ , which represents the surface to be painted.

**Step 4 (Link to Real-Life):** Ask, “If you were to renovate or tile this space, why would knowing area and perimeter matter?”

**Wrap-up:** Facilitate the processing questions.

**Strategy 2: Think-Pair-Share****Instructions:**

1. Present Examples.
2. Let learners *individually* analyze and solve each problem.
3. Learners pair up and compare solutions.
4. They discuss which steps they followed and why, correcting each other if needed.
5. Selected pairs present their answers on the board.
6. Highlights correct procedures, clarifies misconceptions, and reinforces the formula and its application.
7. Facilitate the processing questions.

**Strategy 3: Guided Discovery through Questioning**

1. Begin by posing this statement: “Let’s solve this together. I won’t give you the formula yet — instead, I’ll ask questions and you help me figure it out.”
2. Proceed with guided questioning until the learners verbalize the process.
3. Summarize by asking the processing questions.

**Processing Questions:**

1. How did you decide whether to use area or perimeter for each task in the examples?
2. What clues in the scenario helped you determine which measurement to calculate?
3. Compare the three examples. How is the use of area similar across them? How is the use of perimeter similar?
4. In which situations is perimeter more important than area, and why?
5. Why is it important to know the area when buying materials like colored paper or paint?

- The perimeter is 150 cm, which represents the total length of the welded border.

### **Example 3: Designing a Shot Put Circle**

The athletics coordinator of Agay National High School asks the Grade 11 learners to help repaint the shot-put throwing area in preparation for the school's sports fest. The shot-put circle is circular with a radius of 1.5 meters.

The learners must repaint the entire surface of the circle and mark the boundary line around it with white paint.

#### **Questions:**

1. What quantity should the learners find to know how much area of the circle needs to be repainted?
2. What quantity should they find to determine how long the white boundary line should be?

#### **Expected Answers:**

Repainting → Area

Boundary line → Perimeter (Circumference)

#### **Computation (for checking):**

Given:  $r=1.5\text{m}$

$$\begin{aligned}\text{Area} &= \pi r^2 = 3.14 \times (1.5\text{m})^2 \\ &= 3.14 \times 2.25\text{m}^2 \\ &= 7.065 \text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{Perimeter (Circumference)} &= 2\pi r \\ &= 2 \times 3.14 \times 1.5\text{m} \\ &= 9.42 \text{ m}\end{aligned}$$

#### **Therefore:**

The area of the shot-put circle is  $7.07 \text{ m}^2$ , representing the surface to repaint.

The circumference is  $9.42 \text{ m}$ , representing the total length of the boundary line to mark.

6. Why is it important to know the perimeter or circumference when using borders, ribbons, or boundary lines?



|  | <b>B.2. Discussing the Concept</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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|  | <p style="text-align: center;"><b>Activity B.2: Lesson Discussion</b></p> <p><b>Instructions:</b><br/>Study the information and examples and prepare for class discussion.</p> <p><b>Basic Geometric and Measurement Terms</b><br/> <b>Perimeter</b> – the total distance around the boundary of a closed figure.<br/> <i>Example: The length of the fence needed around a garden.</i><br/> <b>Area</b> – the amount of space a figure covers on a flat surface.<br/> <i>Example: The number of tiles needed to cover a room floor.</i><br/> <b>Length</b> – the longer side of a rectangle or the measurement of how long something is.<br/> <b>Width (or Breadth)</b> – the shorter side of a rectangle; how wide something is.<br/> <b>Height</b> – vertical measure, especially in walls of rooms or fences.<br/> <b>Dimensions</b> – measurements (length, width, height) that describe the size of a figure or object.<br/> <b>Units of Measurement</b> – standard quantities used to measure (e.g., centimeters, meters, square meters, feet, square feet).<br/> <ul style="list-style-type: none"> <li>• <b>Linear units</b> → used for perimeter (cm, m, ft)</li> <li>• <b>Square units</b> → used for area (cm<sup>2</sup>, m<sup>2</sup>, ft<sup>2</sup>)</li> </ul> <b>Shapes Common in Residential Spaces</b><br/> <b>Rectangle</b> – a quadrilateral with opposite sides equal and all angles 90°. Most rooms and gardens are rectangular.<br/> <b>Square</b> – a special rectangle with all sides equal. Example: a small patio or picture frame.<br/> <b>Triangle</b> – a three-sided figure, often seen in roofs or garden corners.<br/> <b>Circle</b> – a round figure, useful in layouts for circular tables, fountains, or round gardens.</p> <p><b>Formulas Commonly Use in Planning and Designing</b><br/> <b>Perimeter of a Rectangle</b> = <math>2 \times (\text{length} + \text{width})</math></p> | <p style="text-align: center;"><b>Activity B.2: Lesson Discussion</b></p> <p>This activity will help learners apply the concepts of area and perimeter in designing and planning residential or functional spaces, demonstrating understanding of how measurement is used in real-life situations.</p> <p>To facilitate the discussion, teacher may select any of the following strategies:</p> <p><i>For Key Terms and Definitions</i><br/> <b>Strategy 1: Math Match-Up</b><br/> <b>Instructions:</b><br/> 1. Prepare sets of cards:<br/> Set A: Key terms<br/> Set B: Formula expressions<br/> Set C: Example<br/> 2. Distribute shuffled cards to small groups.<br/> 3. Groups must match all three correctly: term, formula, and example<br/> 4. After the given time, review answers as a class</p> <p><i>For the Word Problem Examples</i><br/> <b>Strategy 2: Scaffolding</b><br/> <b>Instructions:</b><br/> 1. Solve the first example to demonstrate step by step how to solve a problem. Think aloud while solving. Show how to find the area and perimeter. Write the formulas, substitute values, and explain the meaning (tiles for area, paint for perimeter).<br/> 2. Solve another example with learners' participation. Provide hints or leading questions. Encourage learners to respond while you fill in some steps.<br/> 3. Learners solve remaining problems together, with minimal guidance. The teacher circulates to check and clarify misconceptions.</p> |



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|  | <p style="text-align: center;"><math>P = 2(l + w)</math></p> <p><b>Area of a Rectangle</b> = length <math>\times</math> width<br/> <math>A = lw</math></p> <p><b>Perimeter of a Square</b> = 4 <math>\times</math> side<br/> <math>P = 4s</math></p> <p><b>Area of a Square</b> = side <math>\times</math> side<br/> <math>A = s^2</math></p> <p><b>Perimeter of a Triangle</b> = sum of all sides<br/> <math>P = s_1 + s_2 + s_3</math></p> <p><b>Area of a Triangle</b> = <math>\frac{1}{2} \times</math> base <math>\times</math> height<br/> <math>A = \frac{1}{2}bh</math></p> <p><b>Circumference of a Circle</b> = <math>2 \times \pi \times</math> radius<br/> <math>C = 2\pi r</math>    <b>Area of a Circle</b> = <math>\pi \times</math> radius<sup>2</sup><br/> <math>A = \pi r^2</math></p> <p><b>Practical/Applied Terms</b></p> <p><b>Floor Plan</b> – a simple scaled drawing of a room or house that shows dimensions, layout, and sometimes furniture.</p> <p><b>Layout</b> – arrangement of spaces such as gardens, rooms, or pathways.</p> <p><b>Renovation</b> – process of improving or repairing spaces, where area and perimeter are used to estimate costs (paint, tiles, flooring).</p> <p><b>Budgeting in Home Improvement</b> – using measurements to calculate how much material (tiles, paint, wood, cement) is needed and how much it will cost.</p> <p><b>Accuracy</b> – correctness in measuring and calculating, important to avoid waste in materials or expenses.</p> <p><b>Application</b></p> <p><b>Example 1.</b></p> <p><i>Problem:</i><br/> A circular tabletop has a radius of 70 centimeters. A decorative trim will be placed along the outside edge of the table, and cloth will be used to cover the top.</p> <p><i>Questions:</i><br/> a. How long is the trim in meters?</p> | <p>4. Let learners present their solutions. apply learning individually. Facilitate a short discussion where learners explain the usefulness of area and perimeter in real life.</p> <p>Strategy 3: <b>Peer Teaching</b></p> <ol style="list-style-type: none"> <li>1. Create teaching stations for each term with formulas.</li> <li>2. At each station, prepare: <ul style="list-style-type: none"> <li>o The formula for that term.</li> <li>o An example problem related to the related to the given term.</li> </ul> </li> <li>3. Split the class into small groups and assign each group to a station.</li> <li>4. Each group explains the terms, the formula and the examples.</li> <li>5. After the given time, rotate the groups to the next station.</li> <li>6. When all groups have visited each station, bring the class together to discuss any challenges or solutions.</li> </ol> <p><b>Note:</b><br/> Assess learners understanding after the discussion by asking the questions below.</p> <ol style="list-style-type: none"> <li>1. How do the formulas for area and perimeter help you solve real-life problems such as designing rooms, planning renovations, or preparing for events?</li> <li>2. In which everyday situations outside the classroom do you think you could use what you learned about area and perimeter? Give at least two examples and explain why.</li> <li>3. Why is accuracy in measurement and calculation important when planning home improvement, construction, or cooking activities? What could happen if measurements are incorrect?</li> </ol> |
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b. How many meters of cloth are needed to cover the top of the table?

*Solutions:*

Given: radius of the round tabletop= 70cm

a. Find: Length of the trim (Circumference)

Formula:  $C = 2\pi r$

substitute  $r = 70\text{cm}$

$$C = 2(3.1416) (70\text{cm})$$

$$\mathbf{C = 439.82 \text{ cm}}$$

The trim is about 493.8 cm, convert cm to m  
(100 cm =1m):

$$493.8 \text{ cm} = \frac{1\text{m}}{100\text{cm}} = \frac{493.8 \cancel{\text{cm}}(\text{m})}{100\cancel{\text{cm}}} = 4.938\text{m}$$

**Therefore, the trim length  $\approx 4.94\text{m}$**

b. Cloth covers the area of the circle

Formula:  $A = \pi r^2$ , substitute  $r = 70\text{cm}$

$$A = 3.1416 (70\text{cm})^2$$

$$A = 3.1416 (4900\text{cm}^2)$$

$$A = 15,393.8 \text{ cm}^2$$

Convert  $\text{cm}^2$  to  $\text{m}^2$  ( $1000\text{cm}^2 = 1\text{m}^2$ ):

$$15,393.8 \text{ cm}^2 = \frac{1\text{m}^2}{100\text{cm}^2} = \frac{15,393.8 \cancel{\text{cm}^2}(\text{m}^2)}{100\cancel{\text{cm}^2}} = 1.54\text{m}^2$$

**Cloth area  $\approx 1.54\text{m}^2$**

### **Example 2.**

Problem:

Jean plans to renovate her rectangular living room and design a small garden beside it.

- The living room measures 5 meters by meters.
- The garden beside it is shaped like a rectangle measuring 3 meters by 2 meters.
- Jean wants to put a decorative fence around the garden, and she also plans to install tiles in the living room.

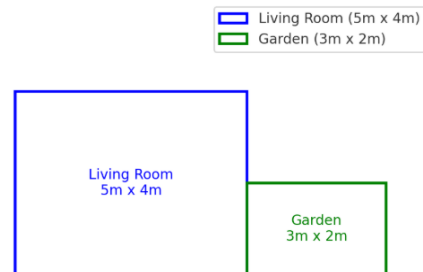
*Task:*

Draw and label the living room and garden floor plan.

*Questions:*

- What is the perimeter of the garden if a fence is to be placed around it?
- What is the area of the living room that needs to be tiled?
- Why is it important to know the area and perimeter when planning renovations and home improvement projects like this?

*Sample Plan Design:*



*Given:*

Living room = 5m by 4m

Garden = 3m by 2m

*Solutions:*

a. Perimeter of the garden (fence length)

Perimeter = 2 (length + width)

$$P = 2(3m + 2m)$$

$$= 2(5m)$$

$$= 10m$$

**Perimeter of the garden to be fenced is 10 meters**

b. Area of the living room (to be tiled)

Area of a rectangle = length(width)

$$A = 5\text{m} (4\text{m})$$

$$A = 20\text{m}^2$$

**Area of the living room that needs to be tiled is 20m<sup>2</sup>**

c. Possible answers why knowing the area and perimeter when planning renovations and home improvement projects are important.

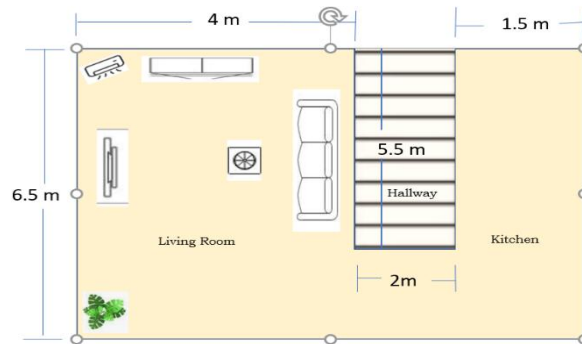
- Accurate material estimates: Area tells you how much flooring/paint/tiles you must buy; perimeter tells you how much skirting, fencing, or trim is needed.
- Cost planning: Multiplying area by unit price gives realistic budgets and prevents overspending or last-minute orders.
- Avoid waste: Buying the right amounts reduces leftover materials and cost.
- Practical design and safety: Ensures furniture fits, walkways are safe, and fences/gates meet space requirements.
- Project timing and labor: Helps estimate how long installation takes and how many workers/tools are needed.

**Example 3.**

*Problem:*

Barbara is getting carpet installed in her entire floor excluding the hallway. Given the floor plan below, answer the following questions:

- a. How many square meters of carpet are needed?
- b. What is the total cost of installing the carpet if the carpet, which measures 5 meters by 2 meters, costs ₱1,000?



*Solutions:*

q. Step 1: Find the total floor area

$$\text{Area}_{\text{floor}} = \text{length (width)}$$

$$\text{Area}_{\text{floor}} = 6.5\text{m} (7.5\text{m})$$

$$\text{Area}_{\text{floor}} = 48.75\text{m}^2$$

Step 2: Find the area of the hallway

$$\text{Area}_{\text{hallway}} = \text{length (width)}$$

$$\text{Area}_{\text{hallway}} = 5.5\text{m}(2\text{m}) = 11\text{m}^2$$

Step 3: Exclude the hallway

$$\text{Carpet area} = 48.75\text{m}^2 - 11\text{m}^2$$

$$\text{Carpet area} = 37.75 \text{ m}^2$$

Answer:

**The carpet needed= 37.75 m<sup>2</sup>**

b. Step 1: Calculate the number of carpet pieces needed

Each carpet covers **10 m<sup>2</sup>**.

$$\text{Number of pieces} = \frac{\text{Total area}}{\text{Area per piece}} = \frac{37.75}{10} = 3.775$$

Since you cannot buy a fraction of a carpet, round up to 4 pieces.

Step 2: Calculate total cost

$$\text{Total cost} = \text{Number of pieces} \times \text{Cost per piece}$$

$$= 4 \times 1000$$

$$= \text{P}4,000$$

**Answer: P4,000**

**Example 4.**

*Problem:*

During a weekend family picnic, Mary is in-charge of grilling chicken and fish. She uses a round grill rack with a radius of 20 cm. She wants to know the cooking area and perimeter (circumference) of the grill rack to plan how much food she can cook at one time.

*Questions:*

- What is the area of the grill rack?
- What is the perimeter (circumference) of the grill rack? (Use  $\pi = 3.14$ )
- If each fish occupies approximately 100 cm<sup>2</sup> and each chicken piece occupies 150 cm<sup>2</sup>, estimate how many fish and chicken pieces can be grilled at a time without overcrowding.
- Why is it important to estimate the number of pieces before starting to grill? How does this help in real-life cooking?

*Given:* radius (r) = 20cm

$$\pi = 3.14$$

*Solutions:*

- a. Find the area.

$$A = \pi r^2$$

$$A = 3.14(20\text{cm})^2$$

$$A = 3.14(400\text{cm}^2)$$

$$A = 1,256\text{cm}^2$$

The cooking area of the grill is 1,256cm<sup>2</sup>

- b. Find the perimeter (circumference)

$$C = 2\pi r$$

$$C = 2(3.14)(20\text{cm})$$

$$C = 125.6\text{cm}$$

- c. Case 1: Estimate number of fish

Each fish occupies 100 cm<sup>2</sup>:

$$\text{Number of fish} = \frac{\text{Grill area}}{\text{Area per fish}} = \frac{1256}{100} \approx 12.56$$

Mary can grill 12 fish at a time (round down to avoid overlap).

Case 2: Estimate number of chicken pieces

Each chicken occupies 150 cm<sup>2</sup>:

$$\text{Number of chicken} = \frac{1256}{150} \approx 8.37$$

Mary can grill 8 chicken pieces at a time.

Case 3: Estimate combined grilling (fish + chicken)

If she wants both on the grill, leaving some space:

- Total area available = 1256 cm<sup>2</sup>
- Assume leaving 10–20% of space empty for proper cooking. Let's leave 20%:

$$\text{Usable area} = 1256 \times 0.8 = 1004.8 \text{ cm}^2$$

- If we grill 6 chicken pieces:  $6 \times 150 = 900 \text{ cm}^2$
- Remaining area for fish:  $1004.8 - 900 = 104.8 \text{ cm}^2 \approx 1 \text{ fish}$

She could grill 6 chicken pieces and 1 fish comfortably.

d. Possible Answer

It is important to estimate the number of fish and chicken pieces before grilling so that Mary can plan how much food fits on the grill without overcrowding. This helps ensure that the food cooks evenly and thoroughly, avoids wasting time by grilling in too many batches, and prevents raw or burnt spots caused by uneven heat. Estimating also helps her manage cooking time efficiently, prepare enough food for everyone, and use her materials (like charcoal or gas) wisely.



### B.3. Developing Mastery

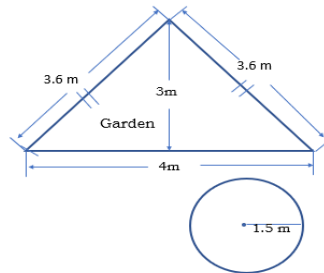
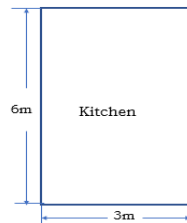
#### Activity B.3: Practice Problems on Area and Perimeter

##### Instructions:

Analyze and solve the given problems.

1. Maria is improving her house by tiling her kitchen floor, building a flower garden, and installing a circular fountain in the yard.

- The kitchen floor is a rectangle measuring 6 meters by 3 meters.
- The flower garden is triangular with a base of 4 meters and a height of 3 meters. The other two sides of the triangle are 3.6 meters each.
- The fountain has a radius of 1.5 meters.



Answer the following questions:

- a. What is the area of the kitchen floor that needs to be tiled?
- b. What is the area of the triangular flower garden?
- c. What is the perimeter of the garden?
- d. What is the area and circumference of the fountain?
- e. Why is it useful for homeowners to know area and perimeter when budgeting for home improvements?

#### Activity B.3: Practice Problems on Area and Perimeter

Practice problems in math help learners understand lessons better by giving them a chance to apply what they have learned. Through regular practice, they become more accurate and confident in solving problems. It also allows them to discover and correct their mistakes, helping them avoid the same errors in the future. By working on different kinds of problems, learners learn to think critically and choose the best strategies to find solutions. Most importantly, practice builds confidence and prepares them for quizzes, exams, and real-life situations where math skills are needed. In short, practice problems are not just about getting the right answers—they are about improving understanding, accuracy, and confidence in learning mathematics.

To facilitate the discussion, teacher may select any of the following strategies:

##### Strategy 1: **Think-Pair-Solve-Share**

Instructions:

1. Give each student a set of problems.
2. Ask them to solve one problem alone (Think).
3. Then, pair them with a seatmate to discuss and solve the next one together (Pair + Solve).
4. Finally, invite volunteers or randomly chosen pairs to share their answers and explain their reasoning to the class (Share).
5. You may use visual aids like posters or digital slides to show each problem.

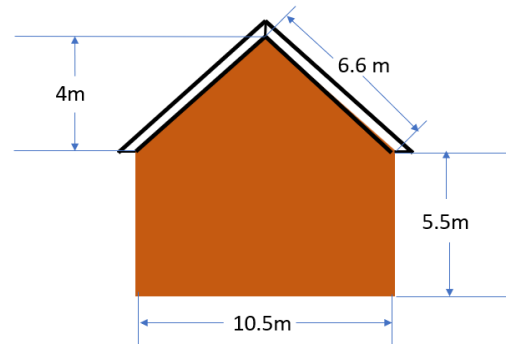
##### Strategy 2: **Color-Coded Stations**

Instructions:

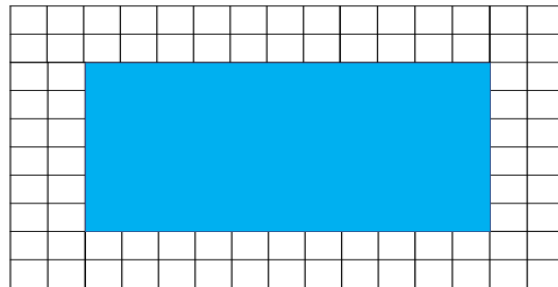
1. Prepare index cards or printed task cards with one problem each (different color for each type of problem).
2. Divide class into small groups of 4–5.

2. Suppose you are painting one side of a house (refer to the illustration below).

- How many square meters of wall will be painted?
- What is the total perimeter of the space to be painted?



3. The diagram below shows the design of a tile border around a rectangular swimming pool that measures 3meters by 5.5meters. Each tile is a square measuring 0.5meter in each side.



#### Questions:

- What is the area of the pool?
- What is the area of the pool and the tiles?
- Using the model above, how many tiles are needed if the pool is 5meters long and 8meters wide?

- Give each group one card to solve together.
- After the given time, say “Pass!” the groups pass their card clockwise to the next group.
- Continue until all groups have solved all cards.
- Do a quick review afterward, clarifying steps and encouraging reflection.

#### Strategy 3: **Interactive Whiteboard Relay**

##### Instructions:

- Divide the class into two teams.
- On the board/projector, display one problem at a time.
- A student from each team races to solve and explain their answer on the board.
- Award points for correct answers and clear explanation.
- Encourage class discussion after each round—ask if anyone 6. solved it differently or spotted errors.

##### Note:

Wrap-up the activity by asking the question:

“Which problem in this activity did you find the most challenging, and what strategies or steps did you use to solve it?”

##### Answer Key

- $18\text{m}^2$ ; b.  $6\text{m}^2$ ; c.  $11.2\text{m}$ ; d.  $A = 7.07\text{m}^2$ ,  $C = 9.4\text{m}$
  - It is useful for homeowners to know the area and perimeter because these measurements help them determine how much material is needed and how much it will cost.
  - Area is needed to calculate flooring, tiles, paint, or soil for gardens.
  - Perimeter is important for fencing, trims, or borders. By knowing these, homeowners can avoid overspending, minimize waste, and make accurate budgets for their renovation projects.
- $78.75\text{m}^2$ ; b.  $34.7\text{m}$
- $16.5\text{m}^2$ ; b.  $16.5\text{m}^2$ ; c. 68 tiles

|                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>C.<br/>Demonstrating<br/>Knowledge and<br/>Skills</b> | <b>C.1. Finding Practical Application</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                          | <p style="text-align: center;"><b>Activity C.1: Home Makeover</b></p> <p><b>Instructions</b></p> <ol style="list-style-type: none"> <li>1. Form Your Team<br/>Work in groups of 3-4. Each group act as a mini home renovation team.</li> <li>2. Choose Your Project<br/>Select one project scenario (or use the one assigned by your teacher): <ul style="list-style-type: none"> <li>• <b>Room renovation</b> (installing tiles, painting walls, or adding molding).</li> <li>• <b>Garden layout</b> (adding a fence, planting grass, or designing flower beds).</li> </ul> </li> <li>3. Draw Your Plan<br/>On manila paper or graphing paper, create a simple floor plan or garden layout. Label the sides with measurements (length and width). Be creative but realistic.</li> <li>4. Do the Calculations <ul style="list-style-type: none"> <li>• Find the perimeter of your layout. (Use this for things like fences, borders, or wall moldings)</li> <li>• Find the area of your layout. (Use this for things like tiles, paint, or grass)</li> <li>• Show your complete solution clearly.</li> </ul> </li> <li>5. Apply the Price List <ul style="list-style-type: none"> <li>• Use the price list provided by your teacher (e.g., cost per meter of fence, cost per square meter of tiles).</li> <li>• Compute the total cost of your renovation project.</li> </ul> </li> <li>6. Prepare Your Presentation <ul style="list-style-type: none"> <li>• Present your group's output to the class.</li> <li>• Be ready to explain: <ul style="list-style-type: none"> <li>○ Your drawing and calculations.</li> <li>○ How perimeter and area were used in your project.</li> <li>○ Why knowing these measurements important in real-life renovations and budgeting?</li> </ul> </li> </ul> </li> </ol> | <p style="text-align: center;"><b>Activity C.1: Home Makeover</b></p> <p>This activity is designed to be completed as group work. Ensure that all learners actively participate and collaborate effectively. This task is an opportunity for them to deepen their understanding and refine their skills, as it will help prepare them for the upcoming performance task. Encourage them to aim for thorough and high-quality work. Provide feedback on their output. In this part you will apply <b>Experiential-Collaborative</b> learning.</p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Introduce the Activity: <ul style="list-style-type: none"> <li>• Explain to students that they will work in groups as mini home renovation teams.</li> <li>• Highlight that the goal is to apply their knowledge of perimeter and area in realistic home or garden renovation projects, including budgeting for materials.</li> <li>• Emphasize teamwork, creativity, and clear presentation of ideas.</li> </ul> </li> <li>2. Form Groups: <ul style="list-style-type: none"> <li>• Organize students into groups of 3–4.</li> <li>• Assign roles if necessary (e.g., designer, calculator, recorder, presenter) to encourage participation from all members.</li> </ul> </li> <li>3. Select a Project: <ul style="list-style-type: none"> <li>• Allow groups to choose a project scenario (room renovation or garden layout) or assign one if needed.</li> <li>• Ensure each group clearly understands the project scope and measurements involved.</li> </ul> </li> <li>4. Guide the Planning Stage: <ul style="list-style-type: none"> <li>• Provide manila paper or graphing paper for groups to create a floor plan or garden layout.</li> <li>• Instruct students to label all sides with measurements (length, width, and any other relevant dimensions).</li> </ul> </li> </ol> |

| Criteria                                                                           | Excellent (5 pts)                                                                                                               | Very Good (4 pts)                                                              | Good (3 pts)                                                                 | Needs Improvement (2 pts)                                                |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>1. Accuracy of Computations (Perimeter and Area)</b>                            | All computations are correct, complete, and clearly shown. Units and formulas are accurate.                                     | Minor errors in computation or labeling but solutions are mostly correct.      | Some errors in formula use or computation that affect accuracy.              | Many incorrect or missing computations; formulas not applied properly.   |
| <b>2. Application of Concepts (Use of Perimeter and Area in Real-life Context)</b> | Demonstrates excellent understanding of how perimeter and area are applied to renovation/budgeting tasks.                       | Shows clear understanding with minor misconceptions.                           | Shows partial understanding of real-life application.                        | Limited or unclear connection between math concepts and real-life tasks. |
| <b>3. Creativity and Layout Design</b>                                             | Layout is neat, realistic, and highly creative. Labels, measurements, and visuals are clear.                                    | Layout is well-made and labeled with slight issues in neatness or proportions. | Layout shows effort but lacks clarity, proportion, or detail.                | Layout is incomplete, messy, or lacks creativity and labeling.           |
| <b>4. Presentation and Explanation</b>                                             | Group presents confidently and clearly explains computations, design choices, and real-life relevance. All members participate. | Presentation is clear with good explanation; most members participate.         | Presentation is understandable but lacks confidence or complete explanation. | Presentation is unclear, incomplete, or participation is minimal.        |
| <b>5. Teamwork and Collaboration</b>                                               | All members contribute actively and show excellent cooperation.                                                                 | Most members contribute with good teamwork.                                    | Some imbalance in participation; limited cooperation.                        | Minimal collaboration; one or two members do most of the work.           |

#### Scoring Guide:

- 18–20 points → Outstanding (Excellent mastery of objectives)
- 14–17 points → Very Satisfactory
- 10–13 points → Satisfactory
- 6–9 points → Needs Improvement
- 1–5 points → Beginning

- Encourage creativity while keeping their designs realistic and practical.
- Circulate and give guidance, checking that measurements make sense and that they understand how to use them in calculations.

#### 5. Facilitate Calculations:

- Remind students to calculate perimeter for items like fences, trims, or borders.
- Remind them to calculate area for items like tiles, paint, or grass.
- Check that students show complete solutions clearly, including formulas, substitutions, and final answers.

#### 6. Apply the Price List:

- Provide the price list for materials (e.g., cost per meter of fence, cost per square meter of tiles).
- Instruct students to compute the total cost of their project using their calculated measurements.
- Encourage them to check their work for accuracy.

#### 7. Prepare and Present:

- Have each group present their project to the class.
- During the presentation, ensure groups explain:
- Their drawing and measurements
- How they used perimeter and area in their calculations
- Why knowing these measurements is important for real-life renovations and budgeting
- Encourage questions from other groups to stimulate discussion.

#### 8. Reflection and Connection:

- After all presentations, lead a class discussion about:
- How perimeter and area affect material planning and costs
- Challenges they faced in measuring, calculating, or budgeting
- How this activity can be applied in their own homes or future.
-

## C.2. Making Generalization

### Activity C.2: Complete the Builder's Guide

#### Instructions:

1. Divide the class into groups with 4-5 members.
2. Each group will receive a worksheet titled "Builder's Guide: Area & Perimeter in Real Life." The guide contains incomplete statements.
3. Base your answers on what you learned in solving problems about room renovations and garden layouts, discuss with your group and write your answers to complete each statement.
4. Make sure your statements explain not just the formulas, but also their usefulness in real life (e.g., for tiling, fencing, painting, budgeting).
5. Be ready to present your group's completed guide and listen to other groups and notice the common ideas.

### Activity C.2: Complete the Builder's Guide

This activity will help learners apply the concepts of area and perimeter in solving real-life problems related to home construction and improvement.

For the summary the best approach to be used is the Reflective Learning. To facilitate this part, you may select follow the strategies below.

#### Instructions:

1. Introduce the Activity
  - Say: "Now that we have practiced solving problems, let's see how well we can apply what we've learned to real-life situations as junior builders!"
  - Distribute the worksheet to each learner.
2. Guide the Class
  - For Part A (*Recall the Basics*), read aloud each item and let learners fill in the blanks individually.
  - Clarify as needed but avoid giving full answers right away—allow learners to recall.
3. Encourage Application
  - For Part B (*Reflect and Apply*), ask learners to:
    - Choose a room or space they know (bedroom, living room, garden).
    - Draw a simple sketch in the provided box.
    - Label length and width, then compute the area and perimeter.
  - Walk around the classroom, checking that sketches are clear and dimensions make sense.
4. Facilitate Reflection
  - For Part C (*Builder's Reflection*), ask learners to complete the open-ended sentence.
  - Prompt deeper thinking by asking:
    - "Why do you think area and perimeter are important in real life?"
    - "How could this knowledge help you or your family in the future?"
5. Class Sharing (5 minutes)

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>C.3. Evaluating Learning</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                 | <p style="text-align: center;"><b>Activity C.3. Assessing Learners Gain</b></p> <p><b>Instructions:</b><br/> Answer the following questions. Show your solutions for computation items and use neat sketches for drawing items.</p> <ol style="list-style-type: none"> <li>1. A rectangular living room has a length of 8 m and a width of 6 m. If tiles cost ₱300 per square meter, how much will it cost to tile the entire floor?</li> <li>2. Mr. Cruz is renovating his backyard. He plans to build a rectangular patio that measures 10 meters long and 6 meters wide. At the center of the patio, he will install a circular garden with a radius of 2 meters. <ol style="list-style-type: none"> <li>a. What is the area of the patio?</li> <li>b. What is the perimeter of the patio?</li> <li>c. What is the area of the circular garden?</li> <li>d. How much space (area) remains in the patio for tiles after placing the circular garden?</li> <li>e. If tiles cost ₱250 per square meter, how much will Mr. Cruz spend on tiling the remaining area of the patio?</li> </ol> </li> <li>3. Maria wants to install curtain rods along the four walls of her rectangular living room. The room measures 6 meters long and 4 meters wide. How many meters of curtain rods does Maria need?</li> <li>4. A family plans to put grass on their rectangular garden that measures 10 meters long and 8 meters wide. What is the total area of the garden that will be covered with grass?</li> </ol> | <p style="text-align: center;"><b>Activity C.3. Assessing Learners Gain</b></p> <p>This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation.</p> <p><b>Answer Key:</b></p> <ol style="list-style-type: none"> <li>1. ₱14,400</li> <li>2. a. 60m<sup>2</sup><br/> b. 32m<br/> c. 12.56m<sup>2</sup><br/> d. 47.44m<sup>2</sup><br/> e. ₱11,860.00</li> <li>3. 20 meters of curtain rods</li> <li>4. The total area of the garden that will be covered with grass is 80 square meters.</li> <li>5. The family will spend ₱6,000 for the tiles.</li> </ol> |



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|                                   | <p>5. A rectangular kitchen has a length of 5 meters and a width of 3 meters. The family wants to put tiles on the floor, and each square meter of tile costs ₱400. How much will they spend for the tiles?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>C.4. Additional Activities</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                   | <p style="text-align: center;"><b>Activity C.4: More Exercise on Area and Perimeter</b></p> <p><b>Instructions:</b><br/>Solve the following problems. Show all steps and include units.</p> <p><b>For Remediation:</b></p> <ol style="list-style-type: none"> <li>A rectangular bedroom measures 4 m in length and 3 m in width.             <ol style="list-style-type: none"> <li>Find the area of the room.</li> <li>Find the perimeter of the room.</li> </ol> </li> <li>A rectangular garden has a length of 6 m and a width of 5 m.             <ol style="list-style-type: none"> <li>Find the area for planting grass.</li> <li>Find the perimeter if you want to build a fence around it.</li> </ol> </li> <li>A wall is 3 m high and 5 m wide, but it has a door measuring 1 m × 2 m.             <ol style="list-style-type: none"> <li>Find the total wall area.</li> <li>Find the paintable area after subtracting the door.</li> </ol> </li> </ol> <p><b>For Enhancement</b></p> <ol style="list-style-type: none"> <li>You are designing a rectangular patio that is 8 m long and 6 m wide. You want to:             <ol style="list-style-type: none"> <li>Install tiles costing ₱250 per m<sup>2</sup>. Find the total cost of tiles.</li> <li>Build a border fence costing ₱100 per meter. Find the total fencing cost.</li> <li>Add 10% extra tiles for wastage. How many m<sup>2</sup> do you need to buy, and what is the new total cost?</li> </ol> </li> <li>A garden consists of a rectangle (10 m × 4 m) with a semicircular flower bed at one end (radius = 2 m).</li> </ol> | <p style="text-align: center;"><b>Activity C.4: More Exercise on Area and Perimeter</b></p> <p>This additional exercise may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning. This activity will help learners master formulas for area and perimeter, apply them to real contexts.</p> <p><b>Answer Key:</b></p> <p>Remediation Activity</p> <ol style="list-style-type: none"> <li>Bedroom 4 m × 3 m             <ol style="list-style-type: none"> <li>Area = <math>4 \times 3 = 12 \text{ m}^2</math></li> <li>Perimeter = <math>2 \times (4 + 3) = 2 \times 7 = 14 \text{ m}</math></li> </ol> </li> <li>Garden 6 m × 5 m             <ol style="list-style-type: none"> <li>Area = <math>6 \times 5 = 30 \text{ m}^2</math></li> <li>Perimeter = <math>2 \times (6 + 5) = 2 \times 11 = 22 \text{ m}</math></li> </ol> </li> <li>Wall 3 m × 5 m with door 1 m × 2 m             <ol style="list-style-type: none"> <li>Total wall area = <math>3 \times 5 = 15 \text{ m}^2</math></li> <li>Paintable area = <math>15 - (1 \times 2) = 15 - 2 = 13 \text{ m}^2</math></li> </ol> </li> </ol> <p>Enhancement Activity</p> <ol style="list-style-type: none"> <li>Patio 8 m × 6 m, tiles ₱250/m<sup>2</sup>, fence ₱100/m             <ol style="list-style-type: none"> <li>Area = <math>8 \times 6 = 48 \text{ m}^2 \rightarrow \text{Tile cost} = 48 \times 250 = \text{₱}12,000</math></li> <li>Perimeter = <math>2 \times (8 + 6) = 2 \times 14 = 28 \text{ m} \rightarrow \text{Fence cost} = 28 \times 100 = \text{₱}2,800</math></li> <li>Add 10% extra tiles: <math>48 \times 1.10 = 52.8 \text{ m}^2 \rightarrow \text{Tile cost} = 52.8 \times 250 = \text{₱}13,200</math></li> </ol> </li> <li>Garden rectangle 10 m × 4 m + semicircle radius 2 m             <ol style="list-style-type: none"> <li>Area: Rectangle = <math>10 \times 4 = 40 \text{ m}^2</math>; Semicircle = <math>\frac{1}{2} \times \pi \times 2^2 = \frac{1}{2} \times 3.14 \times 4 \approx 6.28 \text{ m}^2</math></li> </ol> </li> </ol> <p>Total area <math>\approx 40 + 6.28 = 46.28 \text{ m}^2</math></p> |

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|                                      | <p>a. Calculate the total area of the garden (rectangle + semicircle).</p> <p>b. Calculate the perimeter of the garden (only the outer boundary).</p> <p>c. If soil costs ₱50 per m<sup>2</sup>, how much will you spend to fill the garden?</p> <p>3. You are redesigning a rectangular room 5 m × 4 m with a semicircular window (radius = 1 m).</p> <p>a. Find the floor area for tiling.</p> <p>b. Find the perimeter for molding along the walls (exclude the base of the semicircular window).</p> <p>c. If tiling costs ₱300 per m<sup>2</sup> and molding costs ₱50 per meter, calculate the total cost.</p> | <p>b. Perimeter: Rectangle sides (excluding semicircle base) = 10 + 4 + 4 = 18 m + semicircle circumference = <math>\frac{1}{2} \times 2 \times \pi \times 2 = 6.28 \text{ m} \rightarrow \text{Total} \approx 18 + 6.28 = 24.28 \text{ m}</math></p> <p>c. Soil cost = 46.28 × 50 ≈ ₱2,314</p> <p>3. Room 5 m × 4 m + semicircular window radius 1 m</p> <p>a. Floor area = 5 × 4 = 20 m<sup>2</sup></p> <p>b. Perimeter for molding (walls only): 5 + 4 + 5 + 4 - 1? (exclude semicircle base = 2 × radius = 2 m) → 5 + 4 + 5 + 4 - 2 = 16 m → 16 m</p> <p>c. Tiling cost = 20 × 300 = ₱6,000; Molding cost = 16 × 50 = ₱800; Total cost = 6,000 + 800 = ₱6,800</p> |
| <b>III. CONTENT</b>                  | <p align="center"><b>Lesson 8.3: Calculating the Volume and Capacity and Its Application on Cost Estimation</b></p> <p align="center"><i>(Suggested Time Allotment: 3 hours)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>IV. OBJECTIVES</b>                | <p>At the end of the lesson, learners should be able to:</p> <ol style="list-style-type: none"> <li>1. distinguish between volume and capacity.</li> <li>2. calculate the volume of various 3D shapes and convert between common units of volume and capacity to solve practical problems.</li> <li>3. apply their knowledge of volume and capacity to solve cost estimation problems by first calculating the required amount of a material and then determining the total cost based on the material's unit price</li> </ol>                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>V. PROCEDURES</b>                 | <p><b>ACTIVITIES</b></p> <p>(Instructions for Learners)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p><b>ANNOTATION</b></p> <p>(Instructions for Teachers)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>A. Activating Prior Knowledge</b> | <p><b>Activity A.1. Activating Prior Knowledge</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                      | <p align="center"><b>Activity A.1: Which Container Holds More?</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Your teacher will give you a list of items that need to be stored (both liquids and solids).</li> <li>2. For each item, think of a container that is commonly used to store it (e.g., bottle, box, jar, bag, can, barrel).</li> <li>3. Identify the geometric shape of the container (cylinder, rectangular prism, cube, sphere, cone, etc.).</li> <li>4. Estimate which container can hold more or less. Rank them from smallest to largest capacity.</li> </ol>       | <p align="center"><b>Activity A.1: Which Container Holds More?</b></p> <p>This activity is designed to introduce learners to the concept of capacity and volume by helping them connect everyday containers to geometric shapes and estimate how much they can hold. It encourages learners to start thinking about volume and capacity in real-life contexts, relate common objects to geometric forms, and prepare them for calculating volume using formulas. By engaging with familiar items, learners develop a practical understanding of how different containers store liquids and solids, making the concept of capacity more meaningful and relatable.</p>  |



5. Discuss in pairs or small groups why the chosen container is suitable for the item.

**Guided Questions:**

1. Which containers seem to hold the most? Which hold the least?
2. How does the shape of a container affect how much it can hold?
3. Can some containers hold more than one type of item? Give examples.
4. What measurements do you think you would need to calculate how much each container can hold?

To facilitate this activity, follow the instructions below.

**Instructions:**

1. *Prepare the materials needed:*
  - Pictures or actual objects of containers (bottles, cans, cartons, jars, bags, baskets)
  - Optional: small measuring cups or water for demonstration
  - Worksheet or chart for learners to record their observations
2. *Set the Context*
  - Begin by asking learners: “Think about the containers you use every day. Do they all hold the same amount?”
  - Show examples of containers (real or pictures) such as water bottles, milk cartons, soda cans, rice sacks, tin jars.
3. *Explain the Task*
  - Give learners the list of items (liquids and solids) and ask them to:
    - a. Identify a suitable container for each item.
    - b. Decide the geometric shape of the container.
    - c. Estimate which container holds the most or least and rank them.
  - Emphasize thinking about size and shape, not exact measurement at this point.

Example Items and Containers:

| Item        | Container      | Shape             | Estimated Capacity | Reason for Choice             |
|-------------|----------------|-------------------|--------------------|-------------------------------|
| Water       | Plastic bottle | Cylinder          | Medium             | Easy to carry and pour        |
| Milk        | Carton         | Rectangular prism | Medium-large       | Keeps liquid fresh, stackable |
| Soft drinks | Can            | Cylinder          | Small              | Sealed and portable           |

|         |                 |                              |        |                               |
|---------|-----------------|------------------------------|--------|-------------------------------|
| Rice    | Sack/bag        | Rectangular prism            | Large  | Holds large quantity, durable |
| Cookies | Tin jar         | Cylinder                     | Small  | Keeps cookies fresh           |
| Apples  | Basket          | Rectangular prism / Oval     | Medium | Holds many fruits, ventilated |
| Juice   | Bottle / Carton | Cylinder / Rectangular prism | Medium | Easy to pour, store in fridge |

### 3. Group Work

- Divide learners into pairs or small groups.
- Have them complete the table/chart: item → container → shape → estimated capacity → reason for choice.
- Encourage discussion within groups about why they chose certain containers.

### 4. Class Discussion

- Ask each group to share their choices and rankings.
- Facilitate discussion using the guide questions.

### 5. Connect to Capacity and Volume

- Explain that the activity helps us understand capacity, which is the amount a container can hold.
- Introduce the idea that later they will calculate volume using formulas, but first, it's important to visualize and compare containers.

### Tips for Teachers:

- Encourage learners to think aloud and explain their reasoning.
- Use real containers if possible; seeing and touching helps learners understand volume.
- Accept multiple reasonable answers; the goal is estimating and connecting shapes to capacity.
- Use this activity as a springboard for hands-on experiments measuring actual volumes with water or rice.
-

| <b>A.2. Establishing the Purpose of the Lesson</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                                                    | <p><b>Activity A.2: How Much Will It Cost to Fill It Up?</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>Read the Scenario Carefully<br/>Your family wants to repaint a room. The paint shop sells: <ul style="list-style-type: none"> <li>4-liter can → ₱650</li> <li>1-liter can → ₱180</li> </ul> You estimate that your room needs about 6 liters of paint.</li> <li>Think and Estimate <ul style="list-style-type: none"> <li>Decide how many cans of each size you would buy to cover 6 liters.</li> <li>Estimate the total cost for each option.</li> </ul> </li> <li>Discuss with a Partner <ul style="list-style-type: none"> <li>Compare your estimates with your partner</li> <li>Decide together which option costs less and why.</li> </ul> </li> <li>Share with the Class <ul style="list-style-type: none"> <li>Be ready to explain your choice: <ul style="list-style-type: none"> <li>How did you estimate the total volume?</li> <li>How did knowing the volume help you estimate the cost?</li> </ul> </li> </ul> </li> <li>Reflection <ul style="list-style-type: none"> <li>Think about other situations in daily life where you need to know the volume of something to estimate costs (e.g., buying water, fuel, or ingredients).</li> </ul> </li> </ol> | <p><b>Activity A.2: How Much Will It Cost to Fill It Up?</b></p> <p>This activity introduces learners to the practical importance of <i>volume and capacity</i> in <b>cost estimation</b>. Learners are given a real-life scenario (repainting a room) and asked to estimate how much paint to buy and the total cost. This encourages critical thinking, discussion, and application of mathematical reasoning in everyday financial decisions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p><b>Materials Needed:</b></p> <ul style="list-style-type: none"> <li>Pictures or actual examples of paint cans labeled with <b>volume and price</b> (e.g., 4-liter can ₱650, 1-liter can ₱180)</li> <li>Calculators (optional)</li> <li>Paper and pens for learners to write their estimates</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p><b>Procedure:</b></p> <ol style="list-style-type: none"> <li><b>Set the Context</b> <ul style="list-style-type: none"> <li>Present the scenario: “Your family wants to repaint a room. The paint shop sells different sizes of paint cans. You need to buy enough paint to cover the walls, but you also want to spend as little as possible.”</li> <li>Show the containers (real or images) with their volumes and prices.</li> </ul> </li> <li><b>Think and Estimate</b> <ul style="list-style-type: none"> <li>Ask learners to individually estimate how many cans of each size they would buy to cover 6 liters of paint.</li> <li>They should calculate the total cost for each option.</li> </ul> </li> <li><b>Pair Discussion</b> <ul style="list-style-type: none"> <li>Have learners pair up to compare their estimates.</li> <li>Encourage them to discuss which option is more cost-effective and why.</li> </ul> </li> <li><b>Class Sharing</b> <ul style="list-style-type: none"> <li>Invite pairs to share their reasoning with the whole class.</li> </ul> </li> </ol> |
|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Highlight different strategies for estimating volume and cost.</li> </ul> <p>5. <b>Facilitate Reflection</b></p> <ul style="list-style-type: none"> <li>Ask guiding questions:           <ul style="list-style-type: none"> <li>“How did knowing the volume help you estimate the cost?”</li> <li>“Can you think of other situations in daily life where volume affects cost?”</li> </ul> </li> </ul> <p>6. <b>Transition to Lesson</b></p> <ul style="list-style-type: none"> <li>Conclude:           <ul style="list-style-type: none"> <li>► “Today, we will learn how to accurately measure and estimate volume and capacity, which helps us make smarter financial decisions in real-life situations.” Then present the performance task</li> </ul> </li> </ul>                                                                                                                                                                                                                                |
| <b>B. Instituting New Knowledge</b> | <b>B.1. Presenting Examples</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                     | <p><b>Activity A.2: Volume, Capacity, and Cost Challenge</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>Work in small groups and review the scenarios provided by your teacher.           <ul style="list-style-type: none"> <li>Water Tank Filling: A household has a cylindrical water tank with a radius of 1.2 meters and a height of 2 meters. Water costs ₱35 per cubic meter. Calculate the volume and total cost to fill the tank.</li> <li>Painting a Wall: A classroom wall measures 6 meters long and 3 meters high, and both sides need painting. One liter of paint covers 10 square meters, and a liter of paint costs ₱450. Calculate the area, amount of paint needed, and total cost.</li> <li>Juice Packaging: Cylindrical bottles with a radius of 3 cm and height of 20 cm will be used to serve juice for 200 students. Each bottle costs ₱15. Determine the volume of one bottle and the total cost for all students.</li> </ul> </li> </ol> | <p><b>Activity A.2: Volume, Capacity, and Cost Challenge</b></p> <p>This activity allows learners to explore real-life examples of volume and capacity and use them to estimate costs, developing practical understanding of measurements in everyday contexts. By applying the concept of volume and capacity to real-life situations, learners connect mathematical calculations to financial decisions, enhancing their practical math skills and critical thinking.</p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>Prepare the materials needed:           <ul style="list-style-type: none"> <li>Pictures or real examples of containers, packaging, or storage items (bottles, tanks, boxes, barrels, swimming pools, paint cans, etc.)</li> <li>Price list for materials or products (e.g., price per liter, per cubic meter, per box)</li> <li>Calculator, ruler, or measuring tape</li> <li>Worksheet or chart for recording measurements, volume calculations, and cost estimates</li> </ul> </li> </ol> |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <ul style="list-style-type: none"> <li>• Buying Rice: Rectangular sacks measuring 0.5 m × 0.3 m × 0.8 m, each costing ₱2,000. Calculate the volume of one sack and the total cost for 5 sacks.</li> <li>• Swimming Pool Filling: A rectangular swimming pool measuring 10 m long, 4 m wide, and 1.5 m deep. Water costs ₱40 per cubic meter. Calculate the volume and total cost to fill the pool</li> </ul> <p>2. For each scenario:</p> <ul style="list-style-type: none"> <li>• Identify the shape of the container or object (cube, cylinder, rectangular prism, etc.).</li> <li>• Calculate the volume or capacity using the correct formula.</li> <li>• Use the given prices to estimate the total cost for filling, purchasing, or packaging.</li> <li>• Discuss any assumptions you make, such as rounding volumes or choosing container sizes.</li> </ul> <p>3. Record your calculations and results in the worksheet or chart provided.</p> <p>4. Each group will present one scenario to the class, explaining your calculations, reasoning, and cost estimation.</p> <p>5. During the presentations, observe and compare how different groups approached the calculations and cost estimation.</p> | <p>2. Divide learners into small groups and provide them with the following scenarios:</p> <p>3. For each scenario, learners should:</p> <p>Identify the shape of the container or object (cube, cylinder, rectangular prism, etc.).</p> <p>Calculate the volume or capacity using appropriate formulas.</p> <p>Estimate the total cost based on the given price.</p> <p>Discuss any assumptions they made (e.g., rounding volumes, container sizes).</p> <p>4. Each group will present one scenario, showing their calculations and explaining how they estimated the cost.</p> <p>5. Wrap-up the activity by facilitating the guide questions.</p> <p><i>Guided Questions:</i></p> <ul style="list-style-type: none"> <li>• How does the shape of a container affect the volume it can hold?</li> <li>• How do you convert volume into cost for real-life items?</li> <li>• Which scenario required the largest capacity or had the highest cost? Why?</li> </ul> <p>6. Conclude by explaining that these scenarios are just some examples of how learners can apply the concepts of volume and capacity in real life, including their use in cost estimation. Emphasize that they will understand further as they go along in the discussion.</p> |
| <b>B.2. Discussing the Concept</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                    | <p style="text-align: center;"><b>Activity B.2:</b><br/><b>Volume and Capacity in Real Life and Cost Estimation</b></p> <p><b>Instructions:</b><br/>To better understand the volume and capacity and its application to cost estimation you, will engage with different activities. Everyone is expected to participate actively.</p> <p>Volume and Capacity: The measurement of the total space occupied by a solid is the volume of a three-dimensional figure. Any object that has length, breadth, and thickness is a three-dimensional figure. The difference between the total</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p style="text-align: center;"><b>Activity B.2:</b><br/><b>Volume and Capacity in Real Life and Cost Estimation</b></p> <p>This activity enables learners to apply formulas for volume and capacity to real-life objects while understanding their practical role in estimating costs. It connects learners' problem-solving experiences to a deeper conceptual understanding of volume, capacity, and cost estimation. Through the activity, learners practice accurate unit conversions, estimate real-world costs based on measured capacities, and link mathematical computations to smart financial and practical decision-making.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

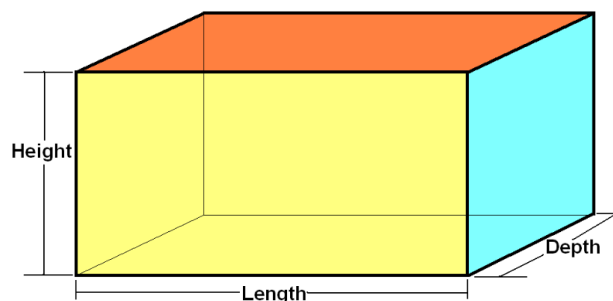
amount of space left inside the hollow body and the space occupied by the body is the volume of a hollow 3-dimensional figure.

Now let's talk about capacity. As we know that there are many hollow objects that we come across in our daily life. Capacity (with reference to volume) may be defined as the maximum amount of matter that an object can hold.

Some of the units for measuring volume are  $\text{cm}^3$ ,  $\text{m}^3$ , liter, milliliter etc.

Let us take the example of a 3D shape called cuboid/rectangular prism, it is a solid bounded by six rectangles and thus has six rectangular faces. The dimensions of the cuboid are as follows: length **l**, depth/width **w**, and height **h**.

Volume of the cuboid is given by the formula:  $V = l \times w \times h$



#### *What is Capacity?*

The hollow objects are filled with air or liquid which take the shape of the container. The total volume of air and water that is accumulated inside a container is the capacity of that container. We can still distinguish between volume and capacity even after many similarities. When dealing with capacity, more concentration is on how much solid, liquid or gas can fit into the container, and it also implies the bearing

To start the discussion you can say : “Now that you know the formulas and conversions, let’s apply them to real-life containers and situations where volume and capacity affect cost. You’ll discover how accurate measurement helps in making wise financial and practical decisions.”

Begin by differentiating volume from capacity.

#### Part 1.

Strategy: Think–Collab–Solve–Share

Instructions:

1. Divide the learners with of 3–4 members, ask them to solve problems collaboratively, then present and explain their reasoning to the class.
2. Visual aids like slides, posters, or physical containers can help illustrate the problems.
3. Facilitate the reflection questions.

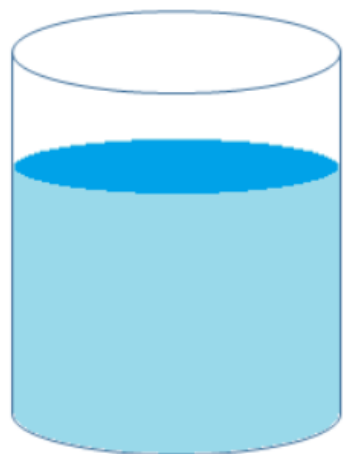
#### Part 2

Strategy: Guided Discussion

Instructions

1. Prepare a simple conversion chart ( $1,000 \text{ cm}^3 = 1 \text{ L}$ ;  $1 \text{ m}^3 = 1,000 \text{ L}$ ) and real-life items such as a water jug (1 L), rice sack, and box.
2. Show two containers labeled 2 L and  $2,000 \text{ cm}^3$ . Ask, “Which holds more? Are they the same?”
3. Ask learners to recall how they converted units in the previous activity (e.g., rice container problem).
4. Display the conversion table and explain the key relations.
5. Measure a small box and calculate its volume step-by-step, converting from  $\text{cm}^3$  to L.
6. Ask learners to look for a partner and answer the guide questions.
7. Provide two short problems for quick conversion and cost estimation practice:
  - a. A small tank measures  $1.5 \text{ m} \times 1 \text{ m} \times 0.8 \text{ m}$ . How many liters can it hold? If water costs ₱35 per kiloliter, what's the cost to fill it?

power of that container. We can measure capacity in  $\text{cm}^3$ ,  $\text{m}^3$ , liters, milliliters, pounds, gallons etc.



### Differences between Volume and Capacity

Both volume and capacity are considered somewhat similar but there are some differences between them that make them different.

- The total amount of any substance which is contained in a particular space is called the volume. The total potential amount of any substance that can be contained in a space is called the capacity of that space.

This can be well understood with the help of a solved example given below.

**Example:** Find the volume and capacity (in liters) of a cylinder whose radius is 7 cm and height is 20 cm.

### Solutions:

Given,  $r = 7 \text{ cm}$  and  $h = 20 \text{ cm}$

We know that,

Volume of cylinder =  $\pi r^2 h$

(Answer:  $1.2 \text{ m}^3 = 1,200 \text{ L} = 1.2 \text{ kL} \times \text{P}35 = \text{P}42$ )

b. A cylindrical milk container has a radius of 7 cm and height of 20 cm. Milk costs  $\text{P}1.20$  per 100 mL. How much will it cost to fill the container?

(Answer:  $V = \pi r^2 h = 3.14 \times 7^2 \times 20 = 3,078 \text{ cm}^3 = 3.078 \text{ L} = 3,078 \text{ mL} \div 100 = 30.78 \times \text{P}1.20 = \text{P}36.94$ )

### Part 3. Linking to Real-Life Cost Estimation

#### Discussion Prompt:

“Now that we can measure and convert volume and capacity accurately, how does this help us estimate and manage costs in real life?”

#### Examples for class discussion:

- Buying bottled water, paint, or gasoline (price per liter)
- Determining the volume of containers for rice or oil to estimate refilling cost
- Calculating water bills based on cubic meters consumed
- Estimating materials (like cement or paint) based on container size and area coverage

#### Encourage learners to reflect:

“Understanding volume and capacity helps us make practical and cost-effective decisions in daily life — whether shopping, budgeting, or managing resources.”

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|  | <p> <math>= (22/7) \times 7 \times 7 \times 20</math><br/> <math>= 22 \times 7 \times 20</math><br/> <math>= 3080 \text{ cm}^3</math><br/> Capacity of the cylinder = <math>3080/1000</math> liters (<math>1000 \text{ cm}^3 = 1</math> liter)<br/> <math>= 3.08</math> liters<br/> Note: We can also represent capacity in terms of <math>\text{cm}^3</math>. Hence, in the above example volume will be numerically equal to capacity </p> <p> Part 1: Volume Reality Check (Problem-Solving Stations)<br/> Directions:<br/> 1. Work in groups of 3–4.<br/> 2. Visit each Problem Station or solve the problems shown by your teacher.<br/> <br/> 3. Compute the volume or capacity, then use it to answer the cost estimation question.<br/> 4. Show your complete solutions and discuss answers within your group.<br/> 5. Be ready to explain how volume or capacity affects the total cost. </p> <p> Problem Stations<br/> Station Scenario Task<br/> 1 A rice container measures <math>40 \text{ cm} \times 30 \text{ cm} \times 50 \text{ cm}</math>. The container will be filled with rice that costs ₱55 per liter. Find the volume in liters and estimate the total cost to fill it.<br/> 2. A cylindrical water tank has a diameter of 2 m and a height of 4 m. Each kiloliter of water costs ₱35. Find the capacity in kiloliters and the total cost to fill the tank.<br/> 3. A balik-bayan box measures <math>0.8 \text{ m} \times 0.6 \text{ m} \times 0.5 \text{ m}</math> and will be filled with liquid soap sold at ₱40 per liter. Find the volume in liters and estimate the total cost if filled completely. </p> <p> Sample Solutions:<br/> Rice Container = <math>40 \times 30 \times 50</math><br/> <math>= 60,000 \text{ cm}^3</math><br/> <math>= 60 \text{ L} \times \text{₱55}</math> </p> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



$$= \text{P}3,300$$

$$\begin{aligned}\text{Cylindrical Tank} &= \pi(1^2)(4) \\ &= 12.57 \text{ m}^3 \\ &= 12.57 \text{ kL} \times \text{P}35 \\ &= \text{P}440.0\end{aligned}$$

$$\begin{aligned}\text{Balik-bayan Box} &= 0.8 \times 0.6 \times 0.5 \\ &= 0.24 \text{ m}^3 \\ &= 240 \text{ L} \times \text{P}40 \\ &= \text{P}9,600\end{aligned}$$

#### Reflection Questions

1. Which formula was easiest to use? Why?
2. What might happen if you forget to convert units correctly?
3. Which of these problems can happen in your home or community?
4. How did knowing the container's capacity help you estimate the total cost?
5. How can this knowledge help in budgeting or making purchasing decisions?

#### Part 2. Concept Connection

Directions:

##### 1. Preparation:

Observe the conversion chart your teacher has prepared:  
Look at the real-life items on the table: a 1 L water jug, a rice sack, and a small box.

##### 2. Hook

Look at the two containers shown: one labeled 2 L, the other 2,000 cm<sup>3</sup>.

Discuss with your partner:

- "Which container holds more?"
- "Are they the same?"

##### 3. Recall

Think about the previous activity (e.g., the rice container problem).

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|  | <p>Share with your partner how you converted the units from <math>\text{cm}^3</math> or <math>\text{m}^3</math> to liters.</p> <p>4. Present</p> <p>Look at the conversion table displayed by the teacher.</p> <p>Take note of the key relationships:</p> <ul style="list-style-type: none"> <li>○ <math>1,000 \text{ cm}^3 = 1 \text{ L}</math></li> <li>○ <math>1 \text{ m}^3 = 1,000 \text{ L}</math></li> </ul> <p>5. Demonstration (Observe as the teacher measures a small box.</p> <ol style="list-style-type: none"> <li>1. Watch the step-by-step calculation: <ul style="list-style-type: none"> <li>○ Measure length <math>\times</math> width <math>\times</math> height in cm</li> <li>○ Compute volume in <math>\text{cm}^3</math></li> <li>○ Convert <math>\text{cm}^3</math> to liters</li> </ul> </li> <li>2. Take note of each step, as you will practice similar calculations.</li> </ol> <p>6. Guide Questions:</p> <ol style="list-style-type: none"> <li>1. Question 1: How is <math>1 \text{ cm}^3</math> related to <math>1 \text{ mL}</math>? <ul style="list-style-type: none"> <li>○ Discuss with your partner. (<i>Answer: They are equal.</i>)</li> </ul> </li> <li>2. Question 2: How is <math>1 \text{ m}^3</math> related to <math>1,000 \text{ L}</math>? <ul style="list-style-type: none"> <li>○ Discuss with your partner. (<i>Answer: <math>1 \text{ m}^3 = 1,000 \text{ L}</math>.</i>)</li> </ul> </li> <li>3. Question 3: Why is it important to use the correct units when buying or estimating items sold by volume? <ul style="list-style-type: none"> <li>○ Share your ideas with the class.</li> </ul> </li> </ol> <p>7. Mini Practice (5–7 minutes)</p> <p>Solve these two problems individually or with your partner:</p> <p>a. Small Tank Problem</p> <ul style="list-style-type: none"> <li>• Tank dimensions: <math>1.5 \text{ m} \times 1 \text{ m} \times 0.8 \text{ m}</math></li> <li>• Water cost: ₱35 per kiloliter</li> <li>• Tasks: <ol style="list-style-type: none"> <li>1. Find the capacity in liters</li> <li>2. Calculate the cost to fill the tank</li> </ol> </li> <li>• (<i>Answer: <math>1.2 \text{ m}^3 = 1,200 \text{ L} = 1.2 \text{ kL} \times \text{₱}35 = \text{₱}42</math></i>)</li> </ul> <p>b. Cylindrical Milk Container</p> <ul style="list-style-type: none"> <li>• Radius = 7 cm, Height = 20 cm</li> </ul> |  |
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|  | <ul style="list-style-type: none"><li>• Milk cost = ₱1.20 per 100 mL</li><li>• Tasks:<ol style="list-style-type: none"><li>1. Find the volume in liters</li><li>2. Calculate the cost to fill the container</li></ol></li><li>• (Answer: <math>V = \pi r^2 h = 3.14 \times 7^2 \times 20 = 3,078 \text{ cm}^3 = 3.078 \text{ L} = 3,078 \text{ mL} \div 100 \times ₱1.20 = ₱36.94</math>)</li></ul> <p>8. Reflection</p> <ul style="list-style-type: none"><li>• Discuss with your partner:<ul style="list-style-type: none"><li>◦ How does knowing volume and capacity help in estimating costs?</li><li>◦ Can you think of other real-life situations where this is useful?</li></ul></li></ul> <p>Part 3. Linking to Real-Life Cost Estimation</p> <p>Directions:</p> <ol style="list-style-type: none"><li>1. Read the discussion prompt: “Now that we can measure and convert volume and capacity accurately, how does this help us estimate and manage costs in real life?”</li><li>2. Think individually for 1–2 minutes about how volume and capacity affect costs in everyday situations.</li><li>3. Share your ideas with a partner or small group, using examples like buying water, paint, or gasoline.</li><li>4. Discuss estimating container volumes for rice, oil, or other goods to plan refills efficiently.</li><li>5. Consider calculating water bills or estimating materials like cement or paint based on container size.</li><li>6. Volunteer to share your insights with the class and listen to new ideas from others.</li><li>7. Reflect briefly on the statement: “Understanding volume and capacity helps us make practical and cost-effective decisions in daily life — whether shopping, budgeting, or managing resources.”</li></ol> |                                                                                                                                                                                                                                                                                     |
|  | <b>B.3. Developing Mastery</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                     |
|  | <p style="text-align: center;"><b>Activity B.3.1: Be a Contractor</b></p> <p>Instructions:</p> <ol style="list-style-type: none"><li>1. Form a group of 4–5 classmates.</li><li>2. Assign your roles: solver, writer, checker, presenter.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p style="text-align: center;"><b>Activity B.3.1: Be a Contractor</b></p> <p>This activity is designed using Collaborative Learning and the Problem-Solving Strategy. By working in groups, learners strengthen their mastery through peer discussion and collective reasoning.</p> |

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|  | <p>3. Work together to solve the problems that will assign to your group.</p> <p>4. The solver solves the problems carefully.</p> <p>5. The checker checks another group's answers.</p> <p>6. The writer prepares your group's output and summary.</p> <p>7. The presenter presents your work to the class.</p> <p><b>Problem Set 1. Home Improvement Project</b></p> <p>1. Filling a Water Drum<br/>A cylindrical water drum has a radius of 0.3 m and a height of 1 m. If water costs ₱50 per cubic meter, how much will it cost to fill the drum completely?</p> <p>2. Painting a Water Tank<br/>A cylindrical water tank has a radius of 0.8 m and height of 2 m. You need to paint the outside surface (including the top but not the bottom). Paint costs ₱100 per square meter. Find the total cost of painting the tank. <i>(Use Surface Area = <math>2\pi r^2 + 2\pi rh</math>)</i></p> <p>3. Constructing a Rectangular Fish Pond<br/>A rectangular pond is 4 m long, 3 m wide, and 1.2 m deep. It will be lined with cement 5 cm thick. Cement costs ₱2,000 per cubic meter.<br/>Compute:<br/>a. The volume of cement needed.<br/>b. The total cost.</p> <p><b>Problem Set 2. School Facility Project</b></p> <p>1. Lower Box Volume<br/>A rectangular flower box measures 1.2 m long, 0.5 m wide, and 0.4 m deep. If soil costs ₱300 per cubic meter, find the total cost to fill one box.</p> <p>2. Rainwater Collector<br/>A cylindrical drum with a radius of 0.4 m and height of 1.2 m is used to collect rainwater. If the drum costs ₱250 per cubic meter of material, estimate the total cost.</p> <p>3. Cylindrical Water Tank with Dome<br/>A tank has a cylindrical base (radius = 1.5 m, height = 3 m) topped with a hemispherical dome. Material costs ₱400 per cubic meter. Find the total volume and cost of the tank.</p> | <p>Instructions:</p> <ol style="list-style-type: none"> <li>Form groups: Divide the class into groups of 4–5 members each.</li> <li>Assign roles: Ensure each group assigns the roles of Solver, Writer, Checker, and Presenter.</li> <li>Distribute problems: Provide each group with a set of problems to analyze and solve collaboratively.</li> <li>Facilitate problem-solving: <ul style="list-style-type: none"> <li>The Solver leads the group in analyzing and solving the problems.</li> <li>The Checker reviews solutions from another group to ensure correctness.</li> <li>The Writer organizes the group's answers and prepares materials for presentation.</li> <li>The Presenter shares the group's work with the class.</li> </ul> </li> <li>Encourage discussion and feedback: During presentations, prompt students to give constructive feedback and help clarify any misconceptions.</li> <li>Guide reflection: Conclude the activity by asking reflective questions to reinforce learning and connect it to real-life applications. <ul style="list-style-type: none"> <li>What strategies did your group use to solve the problems effectively?</li> <li>How did working in groups help you understand the concepts better?</li> <li>What challenges did your group encounter, and how did you overcome them?</li> <li>How did reviewing another group's work (as a Checker) improve your understanding?</li> <li>How can you apply what you learned today to real-life situations?</li> </ul> </li> </ol> <p>Answer Key</p> <p>Problem Set 1: Home Improvement Project</p> <p>1. ₱14.14</p> |
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|                                                          | <p>Problem Set 3. <b>Business Project</b></p> <ol style="list-style-type: none"> <li>Ice Cream Container<br/>A cylindrical container has a radius of 7 cm and a height of 10 cm. If 1 liter of ice cream costs ₱250, find the cost to fill 10 containers. (<math>1,000\text{ cm}^3 = 1\text{ liter}</math>)</li> <li>Milk Carton Production<br/>A rectangular milk box measures 8 cm × 6 cm × 15 cm. If each carton costs ₱0.30 per 100 cm<sup>3</sup> to produce, find the cost of producing 20 cartons.</li> <li>Water Delivery Truck<br/>A cylindrical water tank (radius = 0.6 m, length = 3 m) delivers water at ₱0.15 per liter.<br/>How much will the driver earn per trip if the tank is full?<br/>(<math>1\text{ m}^3 = 1,000\text{ L}</math>)</li> </ol> <p>Problem Set 4. <b>Construction and Design Project</b></p> <ol style="list-style-type: none"> <li>Wooden Box Storage<br/>A rectangular wooden box measures 0.5 m × 0.4 m × 0.3 m. Wood costs ₱1,200 per cubic meter. Find the cost of wood to make one box.</li> <li>Concrete Pillars<br/>A house needs 4 cylindrical pillars, each with a radius of 0.25 m and a height of 3 m. Concrete costs ₱2,500 per cubic meter. Find the total concrete cost for all pillars.</li> <li>Rectangular Pool with Semi-Circular End<br/>A swimming pool is 8 m long, 4 m wide, and 1.5 m deep, with a semicircular end (radius = 2 m). Water costs ₱35 per cubic meter. Find the total cost to fill the pool.</li> </ol> | <ol style="list-style-type: none"> <li>₱1,407.40</li> <li>₱4,080</li> <li>a. <math>2.04\text{ m}^3</math><br/>b. ₱4,080</li> </ol> <p>Problem Set 2: School Facility Project</p> <ol style="list-style-type: none"> <li>₱72</li> <li>₱150.80</li> <li>Volume = <math>28.275\text{ m}^3</math>; Cost ≈ ₱11,310</li> </ol> <p>Problem Set 3: Business Project</p> <ol style="list-style-type: none"> <li>₱3,847.50</li> <li>₱43.20</li> <li>₱508.94</li> </ol> <p>Problem Set 4: Construction and Design Project</p> <ol style="list-style-type: none"> <li>₱72</li> <li>₱5,890</li> <li>₱2,010.88</li> </ol> |
| <b>C.<br/>Demonstrating<br/>Knowledge and<br/>Skills</b> | <b>C.1. Finding Practical Application</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                          | <p><b>Activity C.1: Mini Measurement Challenge</b></p> <p><b>Materials Needed:</b></p> <ul style="list-style-type: none"> <li>Measuring tape, ruler, or meter stick</li> <li>Calculator or mobile calculator app</li> <li>Notebook or data sheet</li> <li>Local prices of materials (water, paint, tiles, or containers)</li> <li>Conversion chart (<math>\text{cm}^3 \rightarrow \text{L}</math>, <math>\text{m}^3 \rightarrow \text{L}</math>, etc.)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p><b>Activity C.1: Mini Measurement Challenge</b></p> <p>Mini Measurement Challenge serves as a foundational activity leading to the performance task “Project Bahay at Paaralan.” It allows learners to apply their understanding of volume and capacity in simple, everyday situations before engaging in a more comprehensive, real-world project that integrates measurement, computation, and cost estimation.</p> <p>This task employs Experiential Learning where learners directly connect classroom concepts to real-life objects in</p>                                                          |

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|                                   | <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Select a small container or space (e.g., a water drum, box, bottle, or small tabletop area).</li> <li>2. Measure its dimensions—length, width, and height (or radius and height if cylindrical).</li> <li>3. Compute the volume using the correct formula:</li> <li>4. Convert the volume to an appropriate unit for capacity (e.g., liters).</li> <li>5. Estimate the cost of filling or covering it using local material prices (e.g., water, paint, or storage contents).</li> <li>6. Record your work clearly in your notebook or data sheet.</li> <li>7. Sketch the object or space with labeled dimensions to visualize your calculations.</li> <li>8. Reflection Questions: <ul style="list-style-type: none"> <li>• How does knowing volume and capacity help in planning costs?</li> <li>• What challenges did you encounter in measuring or computing?</li> <li>• How would your estimate change if the material price increases?</li> </ul> </li> </ol> | <p>their homes or school. By estimating, measuring, and computing, they see how mathematics applies to decisions like choosing the right container or understanding storage space.</p> <p>The Contextualized Teaching Strategy makes the activity authentic, ensuring learners appreciate the relevance of volume and capacity in daily life. Sharing results further encourages communication skills and critical thinking, promoting holistic application of knowledge.</p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. <b>Preparation:</b> Provide learners with sample prices for materials like water, paint, rice, or other items relevant locally.</li> <li>2. <b>Guidance:</b> Monitor proper measurement, correct formula use, unit conversion, and cost calculation.</li> <li>3. <b>Time:</b> 30–40 minutes for measuring, calculating, and estimating costs.</li> <li>4. <b>Discussion:</b> Have learners share their estimated costs and discuss how volume and capacity are used in everyday financial decisions.</li> <li>5. <b>Connection to PT:</b> This practice builds skills for “Project Bahay at Paaralan,” where they will measure larger spaces and compute real-life cost estimates.</li> </ol> |
| <b>C.2. Making Generalization</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                   | <p align="center"><b>Activity C.2:</b><br/><b>Exit Ticket: Volume, Capacity, and Cost Estimation</b></p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Read the prompt carefully:<br/>“Name one item at home or in your community whose volume or capacity you can measure and estimate the cost for. Explain how you would calculate it.”</li> <li>2. Write your answer in 2–3 sentences on the provided exit ticket sheet or notebook.</li> <li>3. Include: <ul style="list-style-type: none"> <li>• The item you chose.</li> </ul> </li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p align="center"><b>Activity C.2:</b><br/><b>Exit Ticket: Volume, Capacity, and Cost Estimation</b></p> <p>This activity will check learners’ understanding of volume, capacity, and their application in estimating costs, and to encourage reflection on real-life relevance. The instructions below shall serve as guide in facilitating this activity.</p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Distribute exit ticket sheets or ask learners to use their notebooks.</li> <li>2. Read the prompt aloud and clarify any questions.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|                                 | <ul style="list-style-type: none"> <li>• Its volume or capacity.</li> <li>• A brief calculation or method for estimating cost.</li> </ul> <p>4. Present your output.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>3. Give learners 3–5 minutes to complete their responses.</p> <p>4. Collect the exit tickets and quickly scan for:</p> <ul style="list-style-type: none"> <li>• Understanding of volume and capacity</li> <li>• Ability to convert units correctly</li> <li>• Application of knowledge to cost estimation</li> </ul> <p>5. Use responses to informally assess learning and guide future lessons.</p> <p>6. Optionally, highlight creative or accurate examples in the next class to reinforce real-life application</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>C.3. Evaluating Learning</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                 | <p style="text-align: center;"><b>Activity C.3:</b><br/><b>Assessment on Volume, Capacity, and Cost Estimation</b></p> <p><b>Instructions:</b><br/>Analyze and solve each problem.</p> <p>1. Cylindrical Drum Problem:<br/>A cylindrical drum has a diameter of 1 m and a height of 1.5 m.</p> <ol style="list-style-type: none"> <li>Calculate its capacity in liters</li> <li>If water costs ₱35 per kiloliter, estimate the total cost to fill the drum.</li> </ol> <p>2. Cube Problem:<br/>A cube has an edge of 20 cm.</p> <ol style="list-style-type: none"> <li>Find its volume in liters.</li> <li>If the cube will be filled with rice costing ₱50 per liter, calculate the total cost.</li> </ol> <p>3. Tank Conversion &amp; Reasoning:<br/>A water tank has a volume of 2 m<sup>3</sup>.</p> <ol style="list-style-type: none"> <li>How many liters of water can it hold?</li> <li>Explain why converting from m<sup>3</sup> to liters is important when calculating water costs.</li> </ol> <p>4. Box Volume and Unit Conversion:<br/>A box has a volume of 45,000 cm<sup>3</sup>.</p> <ol style="list-style-type: none"> <li>Convert this volume to liters.</li> </ol> | <p style="text-align: center;"><b>Activity C.3:</b><br/><b>Assessment on Volume, Capacity, and Cost Estimation</b></p> <p>This assessment activity will check how well learners can apply volume, capacity, and cost estimation concepts independently. It combines calculation problems and conceptual questions, reflecting the problem-solving, guided discovery, and real-life applications practiced in the activities. Providing an answer key allows learners to self-check and helps teachers quickly address misconceptions, reinforcing mastery and preparing learners for future assessments.</p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Explain to learners that the quiz assesses their understanding of volume, capacity, and cost estimation and that they should work individually.</li> <li>2. Distribute the quiz sheets or project the questions on the board.</li> <li>3. Give learners 10–15 minutes to complete all 5 items, including calculations and explanations.</li> <li>4. Walk around the classroom to monitor progress and provide guidance without giving direct answers.</li> <li>5. Remind learners to check units and reasoning for each step in their calculations.</li> <li>6. Collect the quizzes or review answers together using the prepared answer key.</li> </ol> |



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|  | <p>b. If a product costs ₱40 per liter, determine the total cost to fill the box.</p> <p>5. Conceptual Understanding &amp; Analysis:<br/>Which statement about cubic meters is correct, and explain your choice:</p> <p>a) <math>1 \text{ m}^3 = 100 \text{ L}</math><br/>b) <math>1 \text{ m}^3 = 1,000 \text{ L}</math><br/>c) <math>1 \text{ m}^3 = 10,000 \text{ L}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>7. Ask learners to explain their reasoning for selected questions, particularly conceptual ones.</p> <p>8. Highlight common mistakes and clarify misconceptions about volume, capacity, conversions, and cost calculations.</p> <p>9. Emphasize how the skills practiced relate to real-life applications like buying water, rice, or other products sold by volume.</p> <p>10. Encourage learners to reflect on which problems were easy or challenging and how they can apply these skills in everyday life.</p> <p><b>Answer Key:</b></p> <p>1. Cylindrical Drum: 1,178 L; Cost = ₱41.23<br/>2. Cube: 8 L; Cost = ₱400<br/>3. Tank: 2,000 L<br/>4. Box: 45 L; Cost = ₱1,800<br/>5. Correct statement: b) <math>1 \text{ m}^3 = 1,000 \text{ L}</math></p>                                                                                                                                                                                            |
|  | <p><b>C.4. Additional Activities</b></p> <p><b>Activity C.4: More Exercises on Volume and Capacity</b></p> <p><b>Instructions:</b><br/>Analyze and solve each problem</p> <p><b>Remediation Set</b></p> <ol style="list-style-type: none"> <li>1. A cylindrical water container has a diameter of 0.8 m and a height of 1 m. Find its capacity in liters.</li> <li>2. A cube has an edge of 15 cm. Find its volume in liters.</li> <li>3. A tank holds <math>1.2 \text{ m}^3</math> of water. How many liters can it hold?</li> <li>4. A box has a volume of <math>30,000 \text{ cm}^3</math>. Convert this to liters.</li> <li>5. Which is correct:             <ol style="list-style-type: none"> <li>a) <math>1 \text{ m}^3 = 500 \text{ L}</math></li> <li>b) <math>1 \text{ m}^3 = 1,000 \text{ L}</math></li> <li>c) <math>1 \text{ m}^3 = 5,000 \text{ L}</math></li> </ol> </li> </ol> <p><b>Enhancement Set</b></p> | <p><b>Activity C.4: More Exercises on Volume and Capacity</b></p> <p>These problem sets are designed to reinforce and extend learners' understanding of volume, capacity, unit conversions, and cost estimation. The remediation set focuses on building confidence and strengthening foundational skills through simpler, straightforward problems. The enhancement set challenges learners to apply their knowledge to multi-step, real-life scenarios, promoting higher-order thinking, reasoning, and practical application. Both sets align with the earlier activities on problem-solving, guided discovery, and real-life cost estimation, allowing learners to consolidate their skills and prepare for independent application.</p> <p><b>Instructions:</b></p> <ol style="list-style-type: none"> <li>1. Explain the purpose: remediation strengthens basic skills, enhancement challenges learners with complex, real-life problems.</li> </ol> |





strengths, recognize challenges, and plan improvements to enhance student learning. The questions provided below may help in reflecting to your teaching performance.

**1. What aspects of your teaching went particularly well?**

- Which part of the lesson engaged students the most?
- What teaching strategies or methods contributed to successful learning?
- How did students respond to your instruction (e.g., participation, performance, behavior)?
- What evidence shows that learning objectives were achieved?
- What personal teaching strengths were evident in this session?

**2. What challenges did you encounter, and how did you address them?**

- What difficulties arose during lesson delivery or classroom management?
- Were there moments when students struggled to understand a concept? Why?
- How did you adjust your strategies or pacing to respond to these challenges?
- What support or resources would have helped you handle the situation more effectively?
- What will you do differently next time to overcome similar obstacles?

**3 What will you do differently next time to improve student learning?**

- Which teaching strategies or approaches would you modify or try in the future?
- How will you better address diverse learning needs in your classroom?
- Are there new resources, activities, or technologies you plan to incorporate?
- What steps can you take to enhance student engagement or participation?
- How will you assess whether these changes improve learning outcomes?

## Unit 8 – SUMMATIVE TEST

### Multiple Choice

Directions: Read each item carefully. Select the letter of the correct answer and write it on your answer sheet. Each item is worth 1 point.

- Which of the following is used to measure the amount of space a solid object occupies?  
A. Area  
B. Capacity  
C. Volume  
D. Perimeter
- What is the formula for the volume of a rectangular prism?  
A.  $V = l + w + h$   
B.  $V = l \times w \times h$   
C.  $V = 2(l + w + h)$   
D.  $V = \pi r^2 h$
- Which of the following terms describes the total length around the outer edge of a shape?  
A. Area  
B. Capacity  
C. Volume  
D. Perimeter
- What do we call the amount of space a figure covers on a flat surface?  
A. Area  
B. Capacity  
C. Height  
D. Width
- How many liters are equal to one cubic meter ( $1 \text{ m}^3$ )?  
A. 100  
B. 1, 000  
C. 10, 000  
D. 1, 000, 000
- Which of the following units is appropriate for measuring the perimeter of a shape?  
A.  $\text{cm}^2$   
B.  $\text{cm}^3$   
C. cm  
D. mL
- Why is estimation important before measuring?  
A. To waste time  
B. To guess randomly  
C. To avoid accuracy  
D. To make reasonable predictions
- What is the main difference between volume and capacity?  
A. Both measure surfaces  
B. They are exactly the same  
C. Capacity is for solids; volume is for liquids  
D. Volume is for solids; capacity is for containers
- What is the equivalent of  $5000 \text{ cm}^3$  in liters?  
A. 0.5 L  
B. 5 L  
C. 50 L  
D. 500 L
- What unit is most appropriate for measuring a classroom's floor area?  
A.  $\text{cm}^2$   
B.  $\text{m}^2$   
C. L  
D.  $\text{m}^3$
- If a cone and a cylinder have the same base and height, how is the cone's volume related to the cylinder's volume?  
A. Three times greater  
B. One-third as much  
C. Twice as much  
D. Equal
- Which object represents a prism?  
A. Ball  
B. Can  
C. Shoe box  
D. Ice cream cone
- How do you find the perimeter of a triangle?  
A. Add all sides  
B. Subtract one side from another  
C. Multiply all sides  
D. Divide all sides
- What is the volume of a rectangular box measuring  $4 \text{ m} \times 3 \text{ m} \times 2 \text{ m}$ ?  
A.  $9 \text{ m}^3$   
B.  $12 \text{ m}^3$   
C.  $18 \text{ m}^3$   
D.  $24 \text{ m}^3$
- What is the volume of a cylindrical tank with a radius of 1 m and a height of 2 m?  
A.  $2\pi \text{ m}^3$   
B.  $3\pi \text{ m}^3$   
C.  $4\pi \text{ m}^3$   
D.  $6\pi \text{ m}^3$
- A painter covers  $10 \text{ m}^2$  per can of paint. If a wall measures  $40 \text{ m}^2$ , how many cans are needed?  
A. 2  
C. 4

- B. 3 D. 5
17. What is the volume of a cube with a side length of 3 m?  
 A.  $6 \text{ m}^3$  C.  $27 \text{ m}^3$   
 B.  $9 \text{ m}^3$  D.  $81 \text{ m}^3$
18. If a 1 L bottle of glue costs ₱15, how much will it cost to fill a 10 L container?  
 A. ₱100 C. ₱150  
 B. ₱120 D. ₱200
19. Which situation requires estimation rather than exact measurement?  
 A. Baking a cake  
 B. Calculating wall paint  
 C. Measuring a door width for installation  
 D. Approximating the distance to the canteen
20. What does the difference between estimated and actual measurements indicate?  
 A. Mass C. Waste cost  
 B. Accuracy D. Temperature
21. A student measured a table as 150 cm but the actual length is 155 cm. What is the percent error?  
 A. 2% C. 4%  
 B. 3% D. 5%
22. Which information is most important before estimating paint cost for a wall?  
 A. Paint label design  
 B. Wall height only  
 C. Wall color and brand  
 D. Area of the wall and paint coverage
23. Why must a construction worker know both area and perimeter when planning floor tiling?  
 A. To make decorations  
 B. To choose paint color  
 C. Wall color and brand  
 D. To compute both tile quantity and border materials
24. If a tank holds  $1.5 \text{ m}^3$  of water and water costs ₱50 per 100 L, what is the total cost?  
 A. ₱450 C. ₱600  
 B. ₱500 D. ₱750
25. How much will it cost to fence a rectangular garden 8 m by 6 m if the fencing costs ₱200 per meter?  
 A. ₱2000 C. ₱3200  
 B. ₱2800 D. ₱4000
26. A baker needs 12 L of milk for 100 cakes. If milk costs ₱90 per liter, what is the total cost?  
 A. ₱900 C. ₱1,200  
 B. ₱1,080 D. ₱1,500
27. A living room measures  $5 \text{ m} \times 4 \text{ m}$  and a garden beside it measures  $3 \text{ m} \times 2 \text{ m}$ . What is the total area to be tiled?  
 A.  $10 \text{ m}^2$  C.  $20 \text{ m}^2$   
 B.  $16 \text{ m}^2$  D.  $26 \text{ m}^2$
28. If a designer measured a cabinet as 2.1 m tall instead of 2.4 m, what error might occur?  
 A. The paint will fade  
 B. The cabinet fits perfectly  
 C. The cabinet becomes too tall  
 D. The cabinet becomes too short
29. How much will you spend to repaint a  $15 \text{ m}^2$  wall if one liter of paint covers  $3 \text{ m}^2$  and costs ₱250?  
 A. ₱750 C. ₱1250  
 B. ₱1000 D. ₱1500

30. Which real-life task best demonstrates the integration of measurement, geometry, and cost estimation?

- A. Reading a book
- B. Building a table
- C. Watching a movie
- D. Listening to music

### **Unit 8: Performance Task**

#### **Project Bahay at Paaralan: Practical Measurement and Cost Planning**

**Goal:**

You will apply geometric and trigonometric concepts to plan, measure, and estimate materials for a practical project in your local environment, demonstrating accuracy, creativity, and real-life problem-solving skills.

**Role:**

You are a project planner and designer tasked to improve a space or create a three-dimensional object, considering measurement, design, and cost estimation.

**Audience:**

Your teacher, classmates, and community members who will see and evaluate your design and planning output.

**Situation:**

You will conduct a mini-measurement project in your school or home, choosing a space or object that can be redesigned, constructed, or organized. You will measure, compute, and estimate materials needed for improvement or construction using local prices.

**Product/Performance:**

1. **Select a project** from the following options:
  - **Project 1: “Ayos Tayo sa Silid!”** – Classroom redesign (flooring, painting, or furniture layout)
  - **Project 2: “Bahay Improvement sa Barangay”** – Home or small community space (kitchen, garden, terrace)
  - **Project 3: “Tindahan sa Eskwelahan”** – Designing a storage or packaging container
2. **Measure** the dimensions accurately (length, width, height, radius) using rulers, tape measures, or meter sticks.
3. **Compute:**
  - Surface area (for painting, covering, or designing materials)
  - Volume and/or capacity (for storage, container design, or filling purposes)
4. **Estimate costs** based on local material prices for paint, tiles, containers, or other materials.
5. **Present** your output in a clear, organized, and visually appealing format (poster, report, or digital presentation) including:
  - Labeled sketches or diagrams
  - Measurement tables
  - Computation and cost estimate
  - Reflection connecting math concepts to real-life application

**Standards:**

By the end performance task, you will **apply geometric and trigonometric concepts to solve real-life measurement problems** such as estimating dimensions, designing, and constructing three-dimensional objects, demonstrating accuracy, creativity, and practical understanding of shapes and volumes.

Scoring Guide:

| Criteria                                                  | Excellent (4)                                                                                                                              | Proficient (3)                                                                 | Developing (2)                                                                        | Beginning (1)                                                        |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Accuracy of Measurements &amp; Computations</b><br>30% | All measurements are precise; surface area, volume, and capacity calculations fully correct with correct formulas and units.               | Minor errors in measurements or computations; formulas mostly correct.         | Some formulas missing or applied incorrectly; measurements partially accurate.        | Major errors or missing computations; measurements mostly incorrect. |
| <b>Application of Geometric Concepts</b><br>25%           | Correctly applies relevant concepts to all aspects of project, including 3D shapes, composite volumes, and real-life measurement problems. | Applies most concepts correctly; minor misconceptions.                         | Applies some concepts but with errors or incomplete understanding.                    | Limited or no application of geometric/trigonometric concepts.       |
| <b>Cost Estimation</b><br>30%                             | Provides detailed and accurate cost estimate using local prices; includes proper unit conversions and total cost computation.              | Cost estimates mostly correct; minor errors in conversion or totals.           | Cost estimate incomplete or contains several errors; conversion missing or incorrect. | Cost estimate missing or largely inaccurate.                         |
| <b>Presentation and Creativity</b><br>10%                 | Project is very well-organized, visually clear, and creative; includes labeled diagrams, sketches, and tables.                             | Organized and clear; some visual elements or labeling incomplete.              | Somewhat organized; diagrams/tables incomplete or unclear.                            | Disorganized; lacks visual support; difficult to understand.         |
| <b>Reflection and Practical Understanding</b><br>5%       | Reflection demonstrates deep insight; clearly connects measurements, volume, and cost to real-life application.                            | Reflection shows basic understanding and some connection to real-life context. | Reflection lacks depth or is only partially connected to the project.                 | Reflection missing, superficial, or irrelevant.                      |